

SRINIVAS



UNIVERSITY

AI CHARGIFY

Submitted to Srinivas University on partial Completion of Sixth Semester

Bachelor of Computer Application

By

Vinay MH (03SU22SD072)

VI Semester BCA

Under the Guidance of

Prof. Anusha S Rai

Faculty

Institute of Computer Sciences and Information Sciences

Srinivas University

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SRINIVAS UNIVERSITY

Srinivas Nagar, Mukka– 574 146, Mangaluru, Karnataka. Phone: 0824-2477456
(Private University Established by Karnataka Govt. ACT No.42 of 2013,
Recognized by UGC, New Delhi & Member of Association of Indian Universities, New Delhi)
Web: www.srinivasuniversity.edu.in, Email: info@srinivasuniversity.edu.in

INSTITUTE OF COMPUTER SCIENCE & INFORMATION SCIENCES

C E R T I F I C A T E

This is to certify that the project work entitled
“.....
.....
.....” carried out by

Mr./Ms.....

(Registration No.....), a bonafide student of

this institution, in partial fulfillment for the award of

Degree **B.C.A** at Srinivas University, City Campus,

Pandeshwara, Mangaluru – 575 001, during the academic year 2022- 2025. It is

also certified that all corrections/suggestions indicated during internal assessment have been incorporated in the Report. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said degree.

Name & Signature of the Guide External Viva
:

Name & Signature of
the Dean

Name of the Examiners

Signature with Date

1.

2

Internship certificate

CERTIFICATE

OF INTERSHIP

Vinay M H

For active and Invaluable participation of Mr. Vinay M H bearing Reg No – 03SU22SD072 BCA Student of Srinivas University Institute of Computer Science and Information Science, has successfully completed an internship programme on Full Stack Python Developer - Intern, starting from December 2024 to April 2025.

DATE : 30/04/2025

INTERN-ID : TC_MA0720425



SIGNATURE



No.729/2, 2nd Floor, 3rd Cross, Near Sai Baba Temple, Thyagaraja Road, K.R. Mohalla, Mysuru - 570024.



DECLARATION

We thus certify that the project work details included in this report satisfy the requirements needed to be awarded a Bachelor of Computer Application degree.

We hereby declare that, under the direction of **Prof. Anusha S Rai**, Professor in the Department of Computer Science and Information Science at Srinivas University in Mangalore, we have started our project work on "AI Chargify."

To the best of our knowledge and belief, the information contained in this project work report has not been submitted by us for consideration for any other degree at this or any other university. Therefore, we hereby declare that this report is our own original work.

Place: Mangalore

VinayMH(03SU22SD072)

Date: 13-05-2025

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Vinay MH

LIST OF FIGURES

Sl no	Fig No	Title
1	3.2	Context of Flow Diagram
2	3.6	DFD level 0
3	3.6.1	DFD level 1(ADMIN)
5	3.6.8	DFD level 1(User)
6	3.6.13	DFD level 1(EV manager)
8	3.6.18	DFD level 1(Technician)
9	3.8	E-R Diagram
10	5.3.1	Structure Chart for Admin
11	5.3.2	Structure Chart For User
12	5.3.3	Structure Chart For EV manager
13	5.3.4	Structure Chart For Technician
14	5.4.1	Admin
15	5.4.2	User
16	5.4.3	EV manager
17	5.4.4	Technician

CONTENTS

- i. Declaration**
- ii. Acknowledgement**
- iii. List Of Figures**

1. Synopsis

CHAPTER-1

- 1.1 Title of the project
- 1.2 Introduction of the project
- 1.3 Objective of the project
- 1.4 Category
- 1.5 Language to be used
- 1.6 Hardware interface
- 1.7 Software interface
- 1.8 Limitations
- 1.9 Future scope

2. Software Requirement Specification

CHAPTER-2

- 2.1 Introduction
- 2.2 Purpose
- 2.3 scope
- 2.4 Definition, Acronyms, Abbreviation
- 2.5 Overview
- 2.6 Overall Description
- 2.7 Productive Perspective
- 2.8 Browser that supports
- 2.9 Product Function
- 2.10 General Constraints
- 2.11 Assumptions and Dependencies
- 2.12 Specific Requirements
 - 2.12.1 software Requirements
 - 2.12.2 Hardware Requirements
 - 2.12.3 Communication Interface
- 2.13 Functional Requirements
- 2.14 Performance Requirements
- 2.15 Design Constraints
- 2.16 System Attributes

- 2.17 Other Requirements
- 2.17.1 Safety Requirements
- 2.17.2 Security Requirements

3. System Design

CHAPTER-3

- 3.1 Introduction
- 3.2 Context flow Design
- 3.3 Data flow Diagram
- 3.4 Rules Regarding DFD Constraints
- 3.5 DFD Symbols
- 3.6 DFD Level 0
 - 3.6.1 DFD Level 1(admin)
 - 3.6.2 DFD Level 2(Area)
 - 3.6.3 DFD Level 2(EV Station)
 - 3.6.4 DFD Level 2(Technician)
 - 3.6.5 DFD Level 2(Service request)
 - 3.6.6 DFD Level 2(View and manage completed orders)
 - 3.6.7 DFD Level 2(View registered users)
 - 3.6.8 DFD Level 1 (user)
 - 3.6.9 DFD Level 2(User Registration)
 - 3.6.10 DFD Level 2(Ev store)
 - 3.6.11 DFD Level 2(Slot booking)
 - 3.6.12 DFD Level 2(User profile)
 - 3.6.13 DFD Level 1(Ev manager)
 - 3.6.14 DFD Level 2(Slot tracker)
 - 3.6.15 DFD Level 2(Service request)
 - 3.6.16 DFD Level 2(Booking)
 - 3.6.17 DFD Level 2(Orders)
 - 3.6.18 DFD Level 1 (Technician)
 - 3.6.19 DFD Level 2(Service Request)
- 3.7 Entityrelationship Diagram
 - 3.7.1 Data Objects
 - 3.7.2 Attributes
 - 3.7.3 Relationships
 - 3.7.4 Cardinality
- 3.8 ER Diagram

4. Database Design

CHAPTER-4

- 4.1 Introduction
- 4.2 Scope
- 4.3 Table Description
 - 4.3.1 Area Table
 - 4.3.2 Book Slots Table
 - 4.3.3 EV Station Table
 - 4.3.4 Login Table
 - 4.3.5 Slot Tracker Table
 - 4.3.6 Technician Table
 - 4.3.7 Technician Request Table
 - 4.3.8 User Register Table

5. Detailed Design

CHAPTER-5

- 5.1 Introduction
- 5.2 Applicable documents
- 5.3 Structure of the software package
 - a. Structure chart for Admin
 - b. Structure chart for User
 - c. Structure chart for EV Manager
 - d. Structure chart for Technician
- 5.4 Modular decomposition of components
 - a. Admin
 - b. User
 - c. EV Manager
 - d. Technician

6. Coding

CHAPTER-6

6.1 Coding

7. TESTING

CHAPTER – 7

7.1 Introduction

7.2 Testing objectives

7.3 Testing steps

7.3.1 Unit testing

7.3.2 Integration testing

7.3.3 Validation

7.3.4 Output testing

7.3.5 User acceptance testing

7.4 Test cases

8. USER INTERFACE

CHAPTER – 8

8.1 Validation

8.2 Screen short

9. USER MANUAL

CHAPTER –9

9.1 Introduction

9.2 Hardware requirements

9.3 Software requirements

10. CONCLUSION

CHAPTER –10

10.1 Conclusion

11. BIBLIOGRAPHY

CHAPTER – 11

11.1 Website reference

12. PLAGIARISM REPORT

CHAPTER – 12

12.1 Plagiarism

1. SYNOPSIS

1. SYNOPSIS

1.1 Title of the project: AIChargify

1.2 Introduction of project:

- In recent years, more people have started using electric vehicles (EVs), which is a positive move toward greener transportation. EVs have many advantages, such as producing fewer emissions and saving energy by capturing power during braking. This makes them more efficient than regular gasoline cars.
- Yet, a common issue for EV owners is finding charging stations. Unlike regular cars that can refuel at almost any gas station, EV drivers often have trouble locating charging spots, especially during long trips or at crowded stations.
- To tackle this issue, we are developing a project that will help. We are creating a platform specifically for EV users. This platform will help them find nearby charging stations and check their availability in real time. It will have a simple, easy-to-understand interface, making the process of finding and using charging spots much easier. This project will lead to smoother travel experiences and less frustration for electric car drivers.

1.3 Objective of project:

- Help EV users plan charging stops efficiently to reduce waiting time.
- Minimize congestion at charging stations through proper visibility of available slots.
- Enhance convenience for EV owners by supporting better route and charging planning.

1.4 Abstract:

- Electric vehicles (EVs) are becoming more common, bringing improvements in car technology. These improvements make cars use energy more efficiently and are better for the environment. However, EV owners face a big challenge: there aren't as many charging stations as there are gas stations for regular cars. This can make it hard to find a place to charge, especially during long trips.
- This project aims to solve this problem by developing a system that helps people find charging stations in real-time. It will help users easily locate charging spots in different areas. The system will give suggestions by checking how busy each station is, helping people avoid crowded places. This will make the charging process more convenient and reduce wait times. Overall, this solution seeks to make using electric vehicles easier and more practical for daily use and long journeys.

1.5 Category:

Web-based application

1.6 Language to be used:

- Frontend : HTML, CSS, JavaScript
- Backend : Python
- Framework : Bootstrap, Django

1.7 Hardware interface:

- Processor : Intel I3 and above
- RAM : 2GB RAM
- Monitor : 15" Color Monitor

1.8 Software interface:

- Browser: Internet Explorer, Google Chrome, Mozilla Firefox, Microsoft edge
- Operating System. : Windows 10
- Developing Tool : VisualStudio Code
- Database : MySQL

1.9 Limitations

- Dependency on Data Accuracy: The system's efficacy may be impacted by inaccurate or outdated data from charging station APIs.
- A steady internet connection is necessary for real-time data retrieval and map loading.
- Scalability: If the current architecture is not optimized, it may experience performance problems when there is a lot of traffic.
- Limited Station Integration: Initially, only a few charging station networks or providers may be supported.

1.10 Future scope:

- Future scope of this project can be planned that to host it on a real server and increase the maximum usage. In Future,
- we also integrate SMS integration and Email so that instead of login and check notification every time we get notification through sms or by email on every interaction,
- we will integrate a payment gateway then they can pay through the payment gateway

2. SOFTWARE REQUIREMENT SPECIFICATIONS

2. SOFTWARE REQUIREMENT SPECIFICATION

This document explains everything about the "AI Chargify" system, including all its functions and specifications. First, there is an overview of the Software Requirements Specification, or SRS, for the application. Then, it details the project's scope, highlighting what the final software will do and what benefits it will offer. Finally, there is a complete overview to help you understand and navigate the document more easily

2.1 Purpose:

AIChargify aims to fix common problems that people encounter with charging electric vehicles (EVs). Many drivers struggle with too few charging stations and finding an open slot. AIChargify uses artificial intelligence to solve these issues. It offers advice in real-time and even predicts which charging stations will be available. This way, drivers can plan their charging stops more effectively. They won't have to spend a lot of time waiting for a spot to charge their vehicles. This makes the EV charging experience much smoother and more convenient.

2.2 Scope

- AIChargify focuses on improving the EV charging experience by:
- Recommending optimal charging slots: Based on location, proximity to the user, station availability, and historical data.
- Predicting charging slot availability: Using AI to forecast when charging stations will be free, helping users avoid peak times.
- Real-time updates: Providing up-to-date information on charging station availability to allow users to make timely decisions

2.3 Definition, Acronyms, Abbreviation

ADMIN	The Administrator
I/O	Input/output
CFD	Context Flow Diagram
OS	Operating System
DFD	Data Flow Diagram
CSS	Cascading style sheet
RAM	Random Access Memory
SQL	Structured Query Language
HTML	Hyper Text Markup Language
SRS	Software Requirement Specifications ³⁹ⁱ
ERD	EntityRelationship Diagram
HTTP	HyperText Transfer Protocol
USER	Register and raise a complaint

24 Overview:

AiChargify makes charging electric vehicles (EVs) much easier with the use of artificial intelligence. It examines data like how people drive their cars, their locations, and which charging stations have open spots. This allows the platform to recommend the best charging times and predict when slots will be available.

These smart features are designed to help EV drivers avoid long waits and reduce worries about running out of battery, ensuring a smoother and more efficient journey. The system is user-friendly, allowing people to find, reserve, and plan their charging stops with real-time data and intelligent predictions. You can access this service anytime, anywhere through a website, making it very convenient for all drivers.

25 Overall description:

This part of the document will provide an overview of the whole system. It will detail the system's role and how it connects with other systems, as well as introduce its main functions. Additionally, it will describe which different types of users will interact with the system and what specific features are available to each user type. Finally, it will cover any limitations and assumptions associated with the system.

26 Productive perspective:

AiChargify is created to work well with the ever-changing world of electric vehicles and their infrastructure. As more people adopt electric vehicles (EVs), having enough charging stations becomes very important. This platform will help by efficiently managing how charging stations are used and making them more accessible. AiChargify plans to link up with the existing EV networks and infrastructure. It aims to provide a simple and effective solution to the issue of finding available EV charging spots. The platform uses artificial intelligence to give smart advice and predictions. This makes it a key tool to enhance the experience of owning an EV and supports our move towards a greener future.

27 Browser That Supports:

- Google chrome
- Microsoft edge
- Internet explorer
- Mozilla
- Firefox

2.8 Product function

- **Smart Charging Suggestions:** The system gives advice on the best times and places to charge your electric vehicle (EV) by considering your preferences and how busy it gets.
- **Predict Future Charging Needs:** By using past data and smart technology, the system can predict when and where people will need to charge their EVs in the future.
- **Real-Time Station Updates:** Stay informed with live updates on which charging stations are available and how long the wait times are.
- **Easy Network Connection:** This app works well with the EV networks that already exist, making it easy for you to charge your vehicle without hassle.
- **Simple App Design:** The app is user-friendly, making it easy to manage your charging sessions and check your charging history.
- **Flexible Pricing Options:** With different pricing, you are encouraged to charge your vehicle during less busy times, helping to ease congestion.
- **Efficient Energy Management:** It ensures energy is used wisely to prevent overloading the power grid, maintaining a sustainable charging process.
- **Find Nearby Services:** While charging, you can discover nearby services like restaurants and shops or plan routes that include stops for charging.
- **Remote Charging Control:** Use the app to start, stop, and monitor charging sessions from any location via your smartphone.
- **Get Customized Notifications:** The app can send you specific notifications about your charging status, any service updates, and available special offers.
- **Secure Your Information:** Your personal data and charging session details are protected with strong security measures to keep everything safe.

2.9 General Constraints:

General constraints include the following:

- The application works only with an internet connection.
- You need to register the first time you want to access the website.
- Only Admins can handle the roles of Manager and Technician.
- Nobody is allowed to alter someone else's account information on this website.
- The system must recover smoothly and keep all data safe if a device stops working.

2.10 Assumptions and Dependencies:

- This system is made for four types of users: admin, EV manager, technician, and customer.
- Admins must be very careful when they update menu items to avoid mistakes.
- The data entered needs to be correct and kept up to date at all times.
- It is expected that any necessary changes to collect and store data will be done within the current application and database.

2.11 Specific Requirements:

External Interface Requirements:

User Interface:

- To make this application useful for maximum reach of users we have created very user-friendly interfaces so that any kind of user with very less computer skill can use this system.
- The system provides a user-friendly GUI to the users.
- An error message is raised when invalid performance occurs

2.11.1 Software Requirements:

- Operating system: Windows.
- Text editor: Sublime Text 3, Visual studio
- Server: Apache
- Database: MySQL
- Browser: Chrome, Mozilla Firefox, or any other browsing application.

2.11.2 Hardware Requirements:

- Processor: Intel core i3 or higher
- RAM – 4GB and above
- Hard Disk Utilization – 4GB and above

2.11.3 Communication Interface:

The communications functions required by this product are LAN connections within the whole company so that the users can interact with each other

2.12 Functional Requirements:

a) Admin:

- Login: The Admin logs into the platform safely by using secure login details.
- Manage EV Stations: The Admin has the ability to add new EV charging stations, modify existing ones, or remove them when necessary.
- Monitor Slot Availability: The Admin can view and monitor which charging slots are occupied or available in real time at all locations.
- Assign Technicians: The Admin assigns specific tasks to technicians according to the maintenance requests that come in.
- Handle User Complaints: The Admin reviews any complaints users have made and responds to them appropriately.
- View Reports: The Admin is able to generate detailed reports that include data on usage statistics, how efficiently slots are used, and predictions based on this data.
- Manage Technicians: The Admin can add new technicians, update current technician profiles, delete them if needed, and track their ongoing activity in the system.

b) User (EV Owner):

- Registration/Login: You can easily create an account and log in safely.
- Search Charging Stations: Find nearby electric vehicle (EV) charging stations by using filters for location, availability, and slot type.
- View Slot Availability: Check in real time which charging slots are open or already used.
- Book Slot: Choose a station and a slot, decide how long you want to charge, and confirm your reservation.
- Manage Bookings: Look at your current and past bookings. You have options to cancel or change a booking if you need to.
- Make Payment: Pay for your charging sessions through a secure payment method, and keep track of your payment status.
- Receive Notifications: Set up alerts for when slots are available, get booking confirmations, and receive reminders.
- Raise Complaints: If you have any issues with charging stations or sessions, you can report them.
- View Charging History: Access detailed records of all your past charging sessions.

- **Manage Profile:** You can update your personal details like your phone number, address, and login information

c) EV Manager:

- **Sign Up/Login:** Create an account and log in securely.
- **Search for Charging Stations:** Find nearby EV charging stations by using filters like where they are located, if they are available, and the type of slot.
- **Check Slot Availability:** Get live updates on which charging slots are open or in use.
- **Reserve a Slot:** Choose a station and slot, decide how long you need it, and confirm your reservation.
- **Manage Your Bookings:** View your current and past bookings. Cancel or change them if needed.
- **Log In/Register:**
- Securely access the platform with a unique username and password.
- **Manage Station Details:**
- Keep station details updated, such as name, contact info, location, and available charging slots.
- **Check Slot Status:**
- Get real-time updates on each charging slot's status—whether it is free, in use, or under maintenance.
- **Control Charging Slots:**
- Add new slots, make updates, or disable them if necessary.
- **View Bookings:**
- Access the schedule of current and upcoming bookings to monitor slot usage.
- **Approve/Change Bookings:**
- Adjust bookings manually if needed, like in emergencies or when there are system conflicts

d) Technician:

- **Login/Register:** Easily enter the platform by using your username and password. This keeps your access secure.
- **View Repair Requests:** Check all the details for each repair request. You will see the location, slot ID, and what's wrong.
- **Update Repair Status:** You can change the status of maintenance requests. Mark them as pending, in progress, or completed based on their current stage.
- **Manage Profile:** Keep your contact details up-to-date, and don't forget to update your profile picture when needed.

- Access Assigned Stations View the list of stations assigned to you. You'll also find directions to reach each station for your maintenance work.

2.13 Performance Requirements:

- Performance requirements define acceptable response times for system functionality
- The load time for user interface screens shall take no longer than two seconds.
- The log in information shall be verified within five seconds.
- Queries shall return results within five seconds

2.14 Design Constraints:

The " **AIChargify** " system shall include two categories of users: **admin (traffic authorities) and general users (citizens)**. The system is a **web-based platform** designed for managing traffic violations and parking enforcement. It integrates **Automated License Plate Recognition (LPR)** technology to detect violations and process complaints efficiently. The system shall be developed using **PYTHON and MySQL databases**

2.15 System Attributes:

- Reliability: The system must keep working well and try to avoid any breaks or mistakes. It should use strong ways to manage data, so it can handle a lot of location data for charging stations in real time.
- Availability: The system needs to be available all the time, 24 hours a day, 7 days a week. This ensures that people can always see which charging slots are open and get the latest information.
- Security: To get into different parts of the system, like user and admin areas, you need to log in securely with a username and a password. Important information about users and charging stations is protected by encryption, which means it is turned into a code to keep it safe.
- Portability: The system should be able to run on major operating systems such as Windows, macOS, and Linux. These systems have to support the PHP environment and the MySQL database that the system uses to work.

2.16 Other Requirements:

2.16.1 Safety Requirements:

- If you forget or lose your password, don't worry. On the main login screen, there's an option to help you recover it. This process uses secure methods to check who you are and keep things safe.
- We store all client information securely in our database. We have strong controls to make sure only authorized people can access this information, preventing anyone without permission from getting to it.
- We will also carefully check the location data for both users and charging stations. This helps ensure that any recommendations made by our AI are accurate and reliable, making it easier to find what you need.

2.16.2 Security Requirements

Security measures are essential to keep the platform safe and accessible only for those permitted to use it. A strong login system will protect user accounts by encoding passwords and confirming user identities. This ensures only legitimate users can access the platform, preventing unauthorized entry. Any information provided by users, such as their location or charging slot choices, must be checked to avoid incorrect data submission, which helps maintain the system's reliability. The data exchanged between the user and the server will be secured using encryption to protect the privacy and security of both the user's personal and location details.

3. SYSTEM DESIGN

3. SYSTEM DESIGN

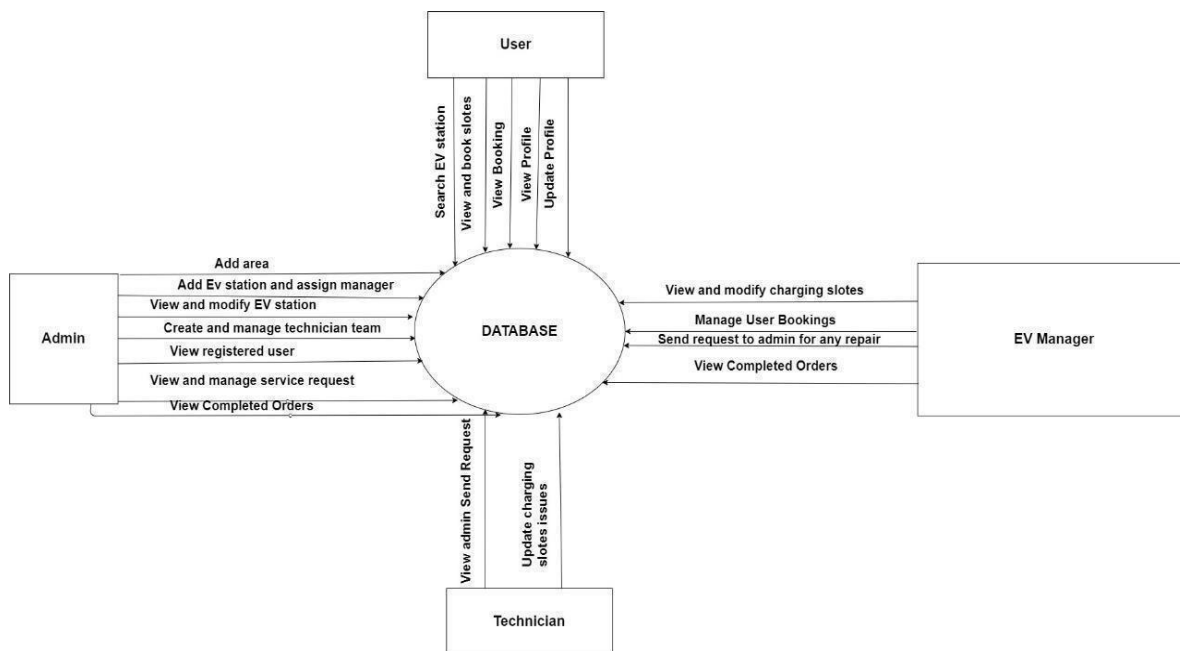
3.1 Introduction:

The design process aims to create a model of a system that helps in building it later. Systems design involves defining the system's architecture, modules, interfaces, and data to meet certain requirements. It focuses on what modules are needed, their details, and how they connect with each other.

Think of systems design as applying systems theory to product development. It uses a step-by-step method to design a system. This can be done bottom-up, starting with details, or top-down, starting with the big picture. Both methods consider all variables related to the system. System design influences the system's main structure, affecting how easy it is to test and change. The result is the architectural design for the software system to be built.

3.2 Context Flow Diagram:

A context flow diagram is a simple type of data flow diagram. It features a single process node that represents the entire system's function in relation to external entities. This diagram treats the entire system as one large process, highlighting all the inputs, outputs, sources, and destinations (also known as sinks). It helps in understanding how the system interacts with external elements by showing these interactions clearly.



3.2Context Dataflow diagram

3.3 Data Flow diagram:

A Data Flow Diagram (DFD) is a visual tool that shows how different parts of a system work and interact. It uses simple symbols to display data movements, processes, where data comes from (sources), where it goes (sinks), and how it is stored.

A DFD is important for understanding how a system functions. It helps you see the main system, the parts that connect with it, and how data and information move around inside it.

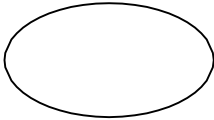
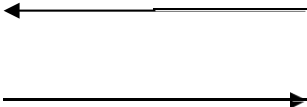
The diagram shows how information travels through the system and undergoes changes from beginning to end. It works like a blueprint that traces the path from input to output.

DFDs are also called bubble charts or Data Flow Graphs. They can be used to show details of a system at different levels. As you look deeper into each level, you get a clearer picture of how data moves and the functions of each part of the system.

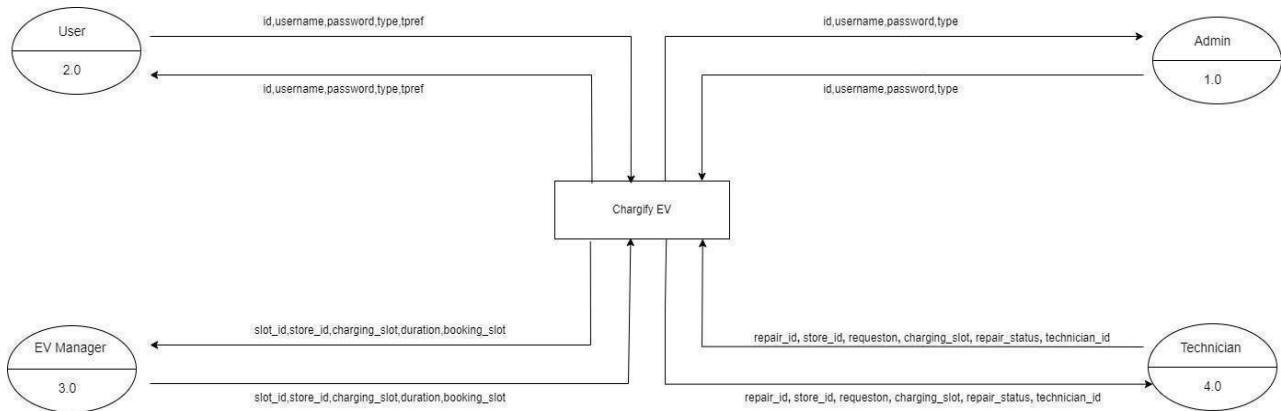
3.4 Rules Regarding DFD construction:

- A process cannot have only outputs.
- A process cannot have only inputs.
- The inputs to a process must be sufficient to produce the outputs from the process.
- All data stores must be connected to at least one process.
- All data stores must be connected to a source or sink.
- A data flow can have only one direction of flow. Multiple data flows to and/or from the same process and data store must be shown by separate arrows.
- If the exact same data flows to two separate arrows, it should be represented by a forked arrow.
- Data cannot flow directly back into the process it has just left. All data flows must be named using a noun phrase.

3.5 DFD Symbols:

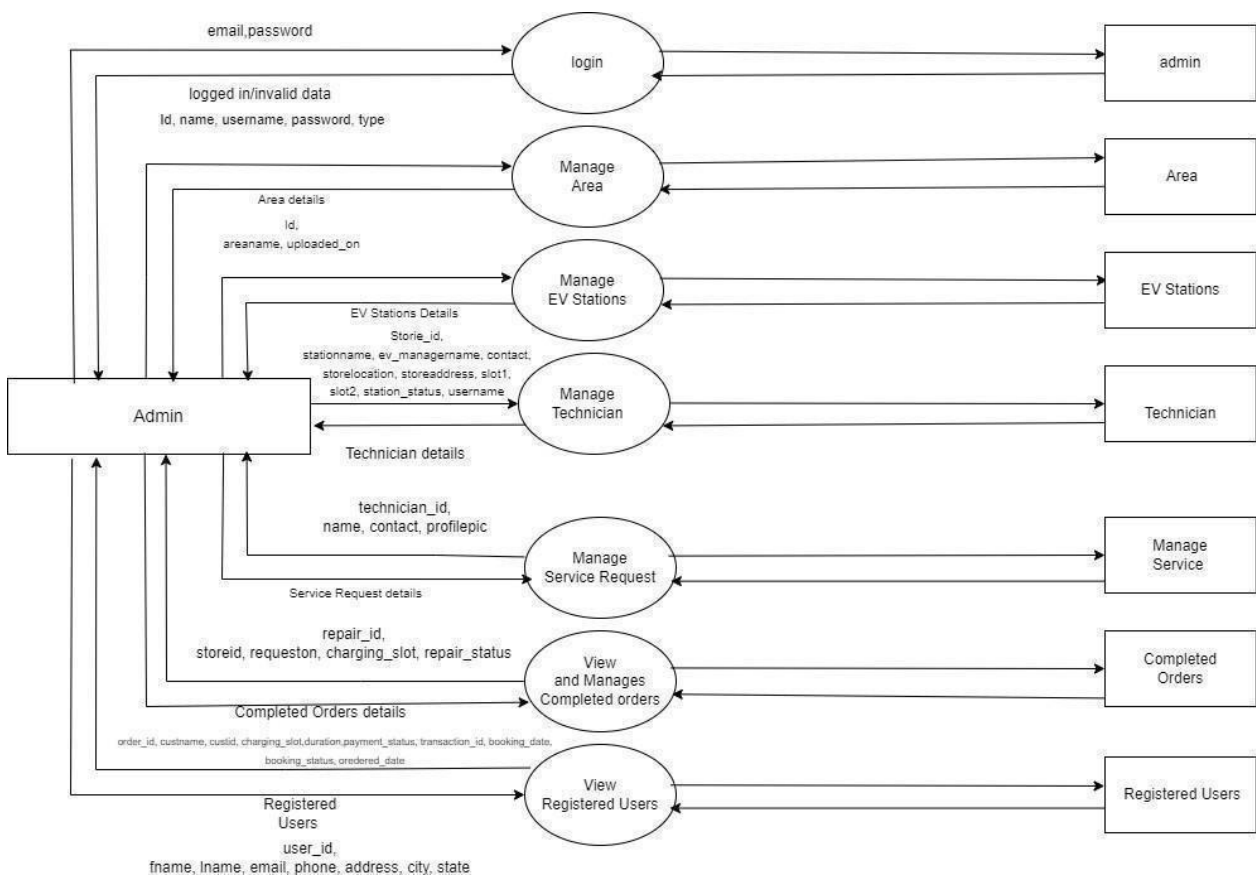
Name	Notation	Description
Process		A process transforms incoming data flow into outgoing data flow. The processes are shown by named circles.
Data store		Data stores are repositories of data in the system. They are sometimes also referred to as files.
Data flows		Data flows are pipelines through which packets of information flow. Label the arrows with the name of the data that moves through it.
External Entity		External entities are objects outside the system with which the system communicates. External Entities are resources and destinations of the system's inputs and outputs

3.6 DFD Level 0:



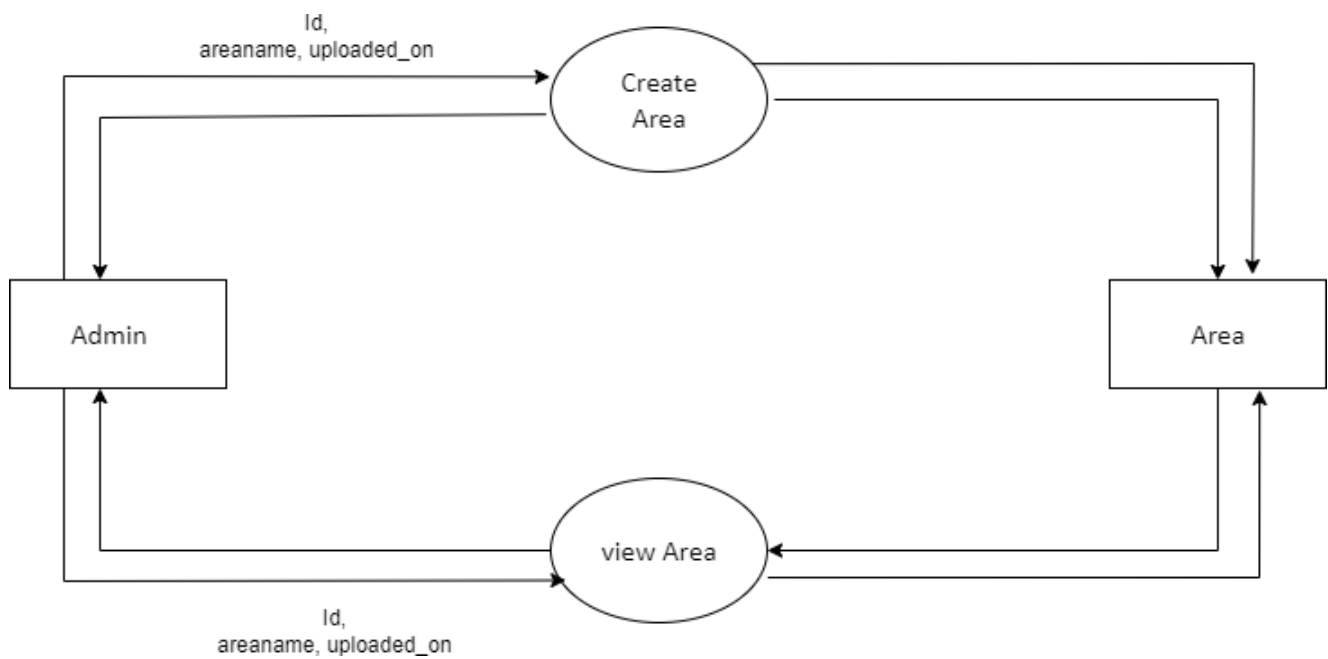
3.6 DFD Level 0

3.6.1 DFD Level 1 (Admin):



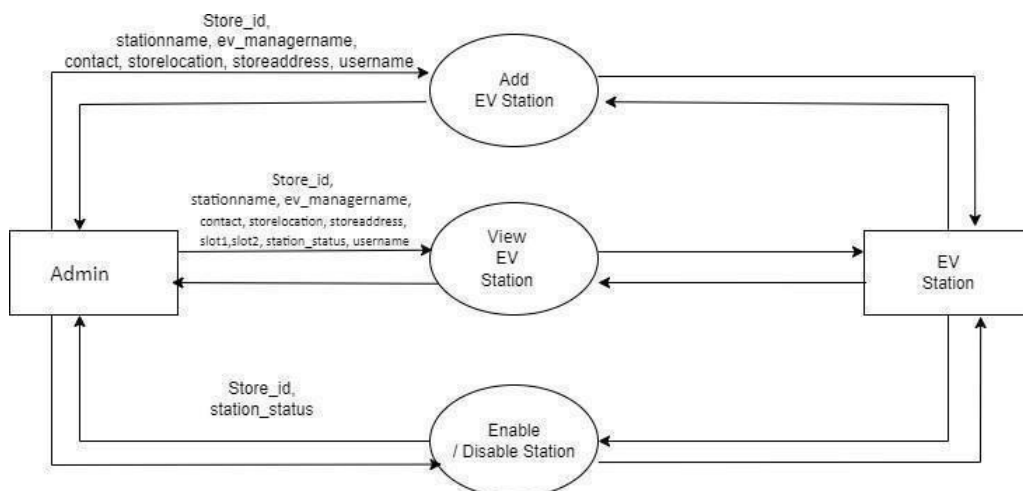
3.6.1 DFD Level 1(Admin)

3.6.2 DFD level 2(Area):



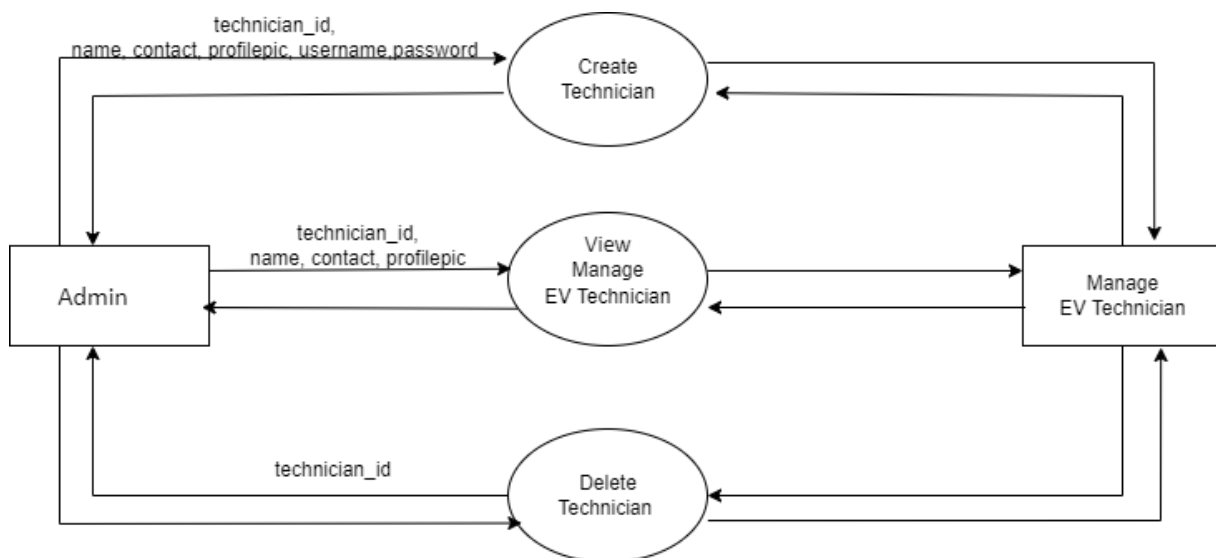
3.6.2 DFD Level 2(Area)

3.6.3 DFD level 2(Ev Station):



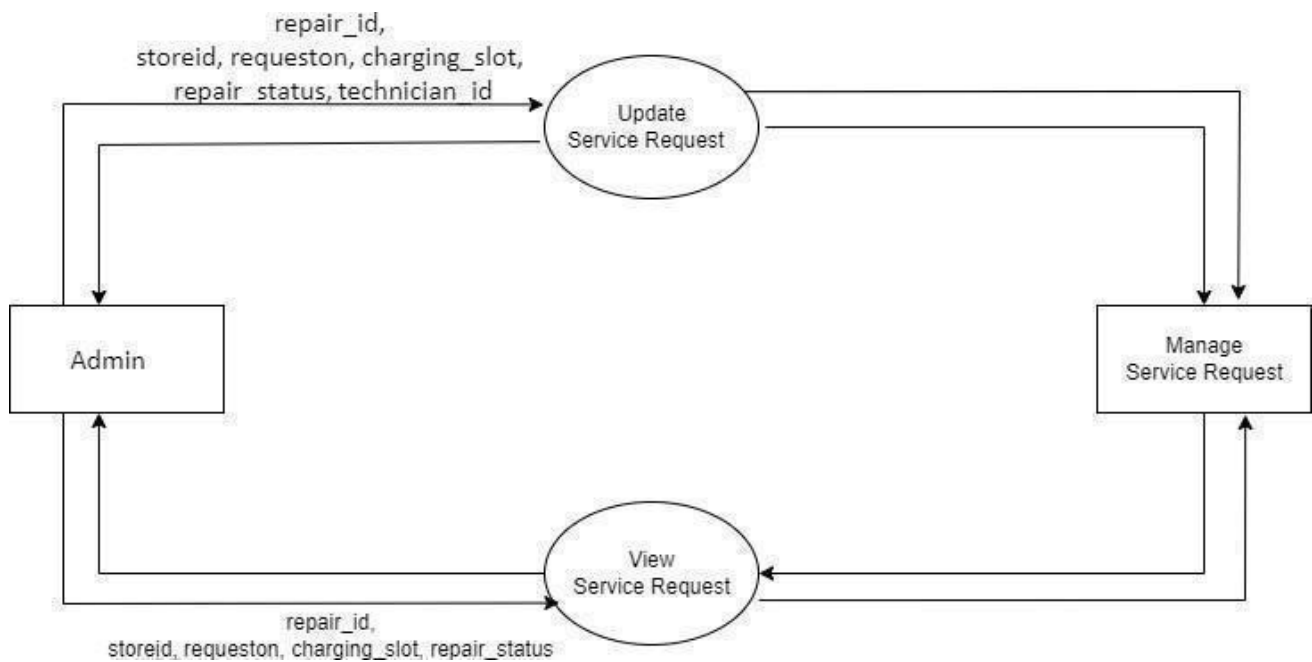
3.6.3 DFD Level 2(EV Station)

3.6.4 DFD level 2(Technician):



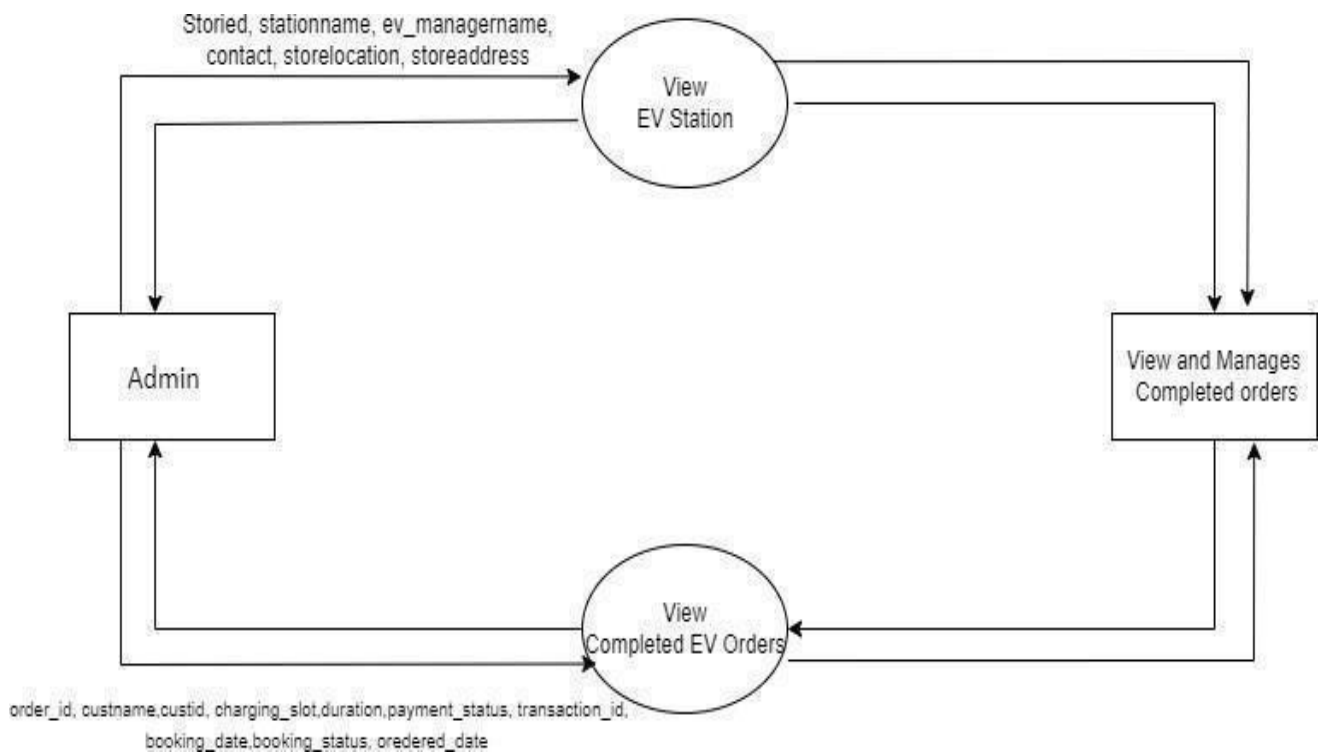
3.6.4 DFD level 2 (Technician)

3.6.5 DFD level 2(Service Request):



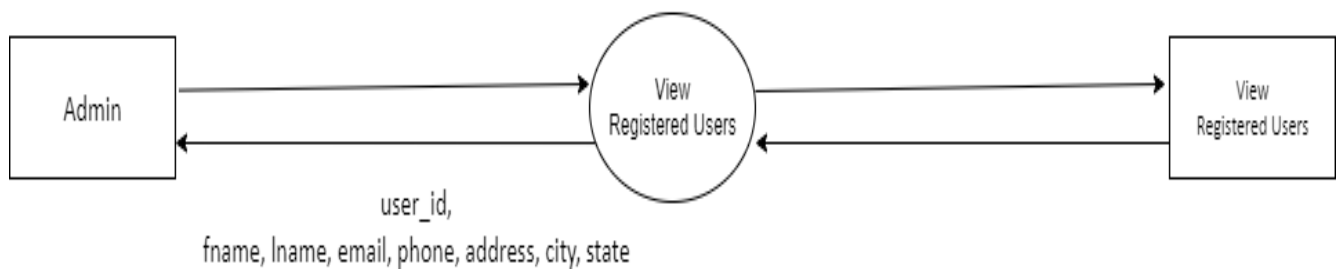
3.6.5 DFD level2 (Service request)

3.6.6 DFD level 2 (View and Manages completed orders):



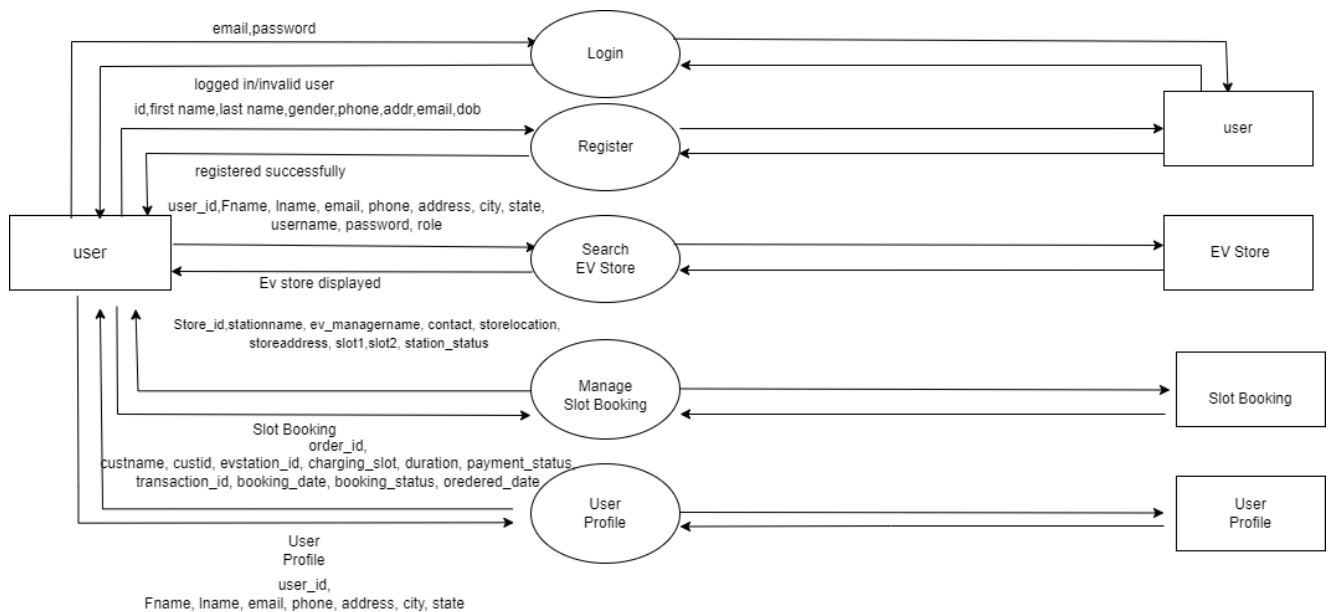
3.6.6 DFD level 2 (View and manages completed orders)

3.6.7 DFD level 2 (Viewregisteredusers):

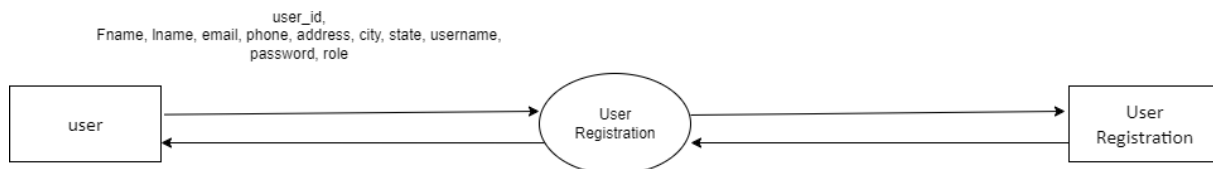


3.6.7 DFD level 1(View registered users)

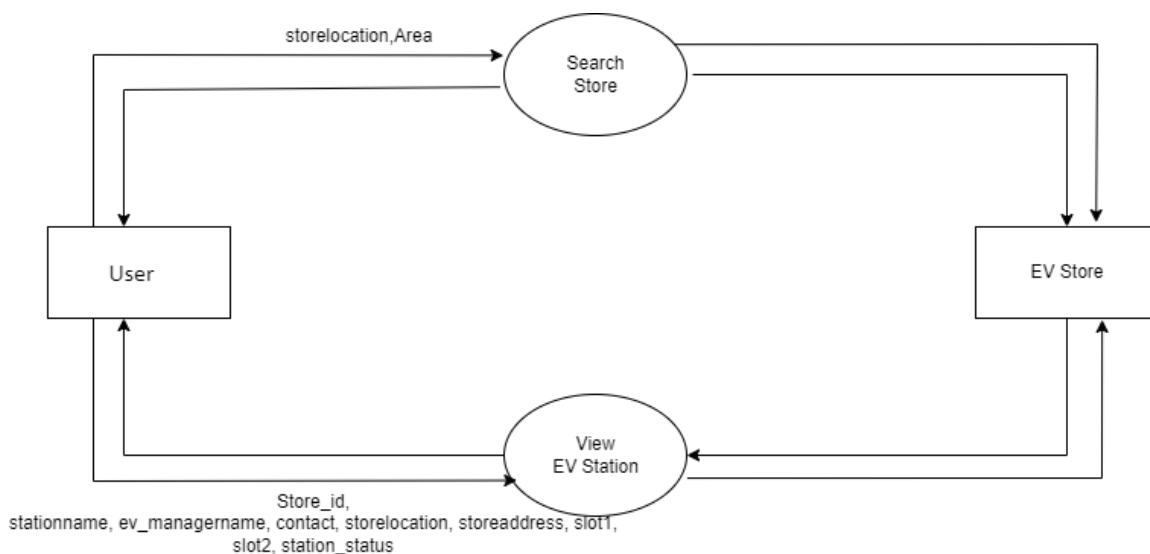
3.6.8 DFD level 1(User):



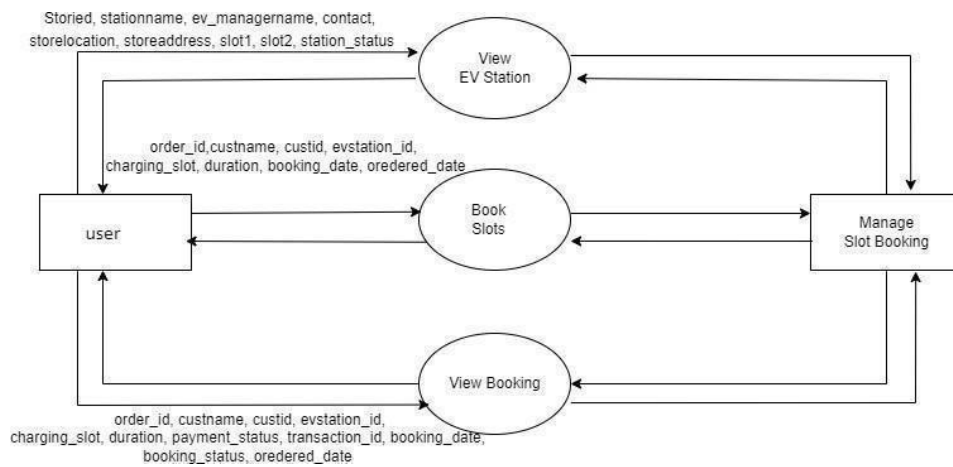
3.6.9 DFD level 2(User Registration):



3.6.10 DFD level 2(EV Store):

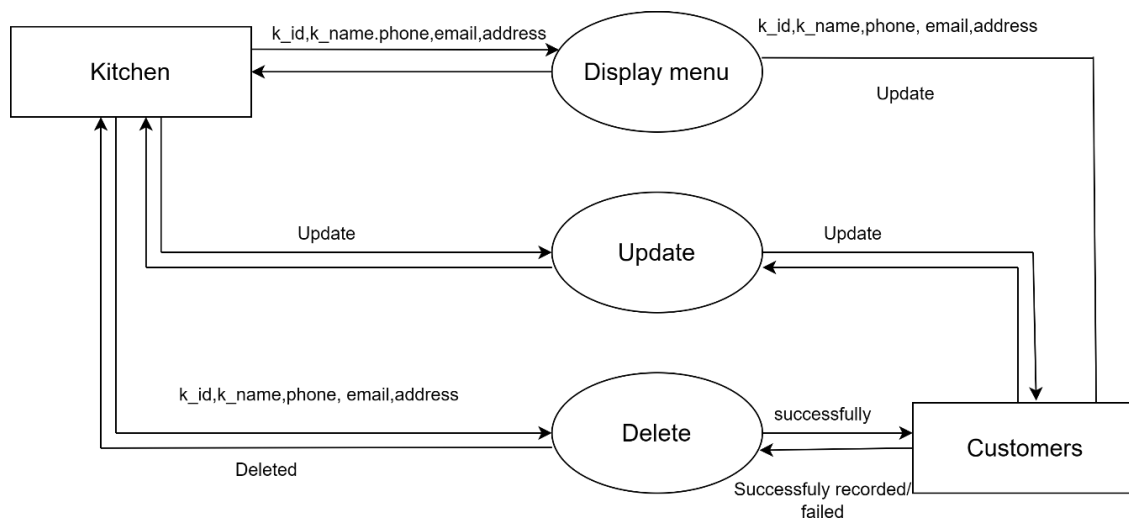


3.6.11 DFD level 2(Slot booking):



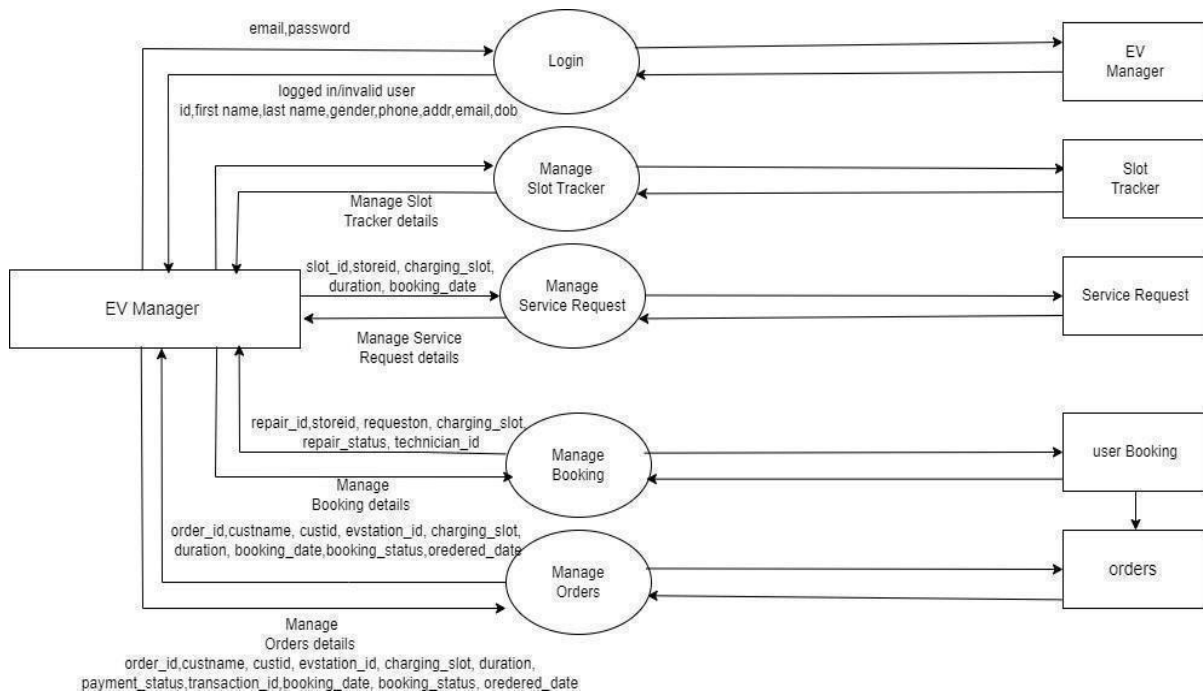
3.6.11 DFD level 2 (Slot booking)

3.6.12 DFD level 2(User Profile):



3.6.12 DFD level 2(User profile)

3.6.13 DFDlevel 1(EV Manager):



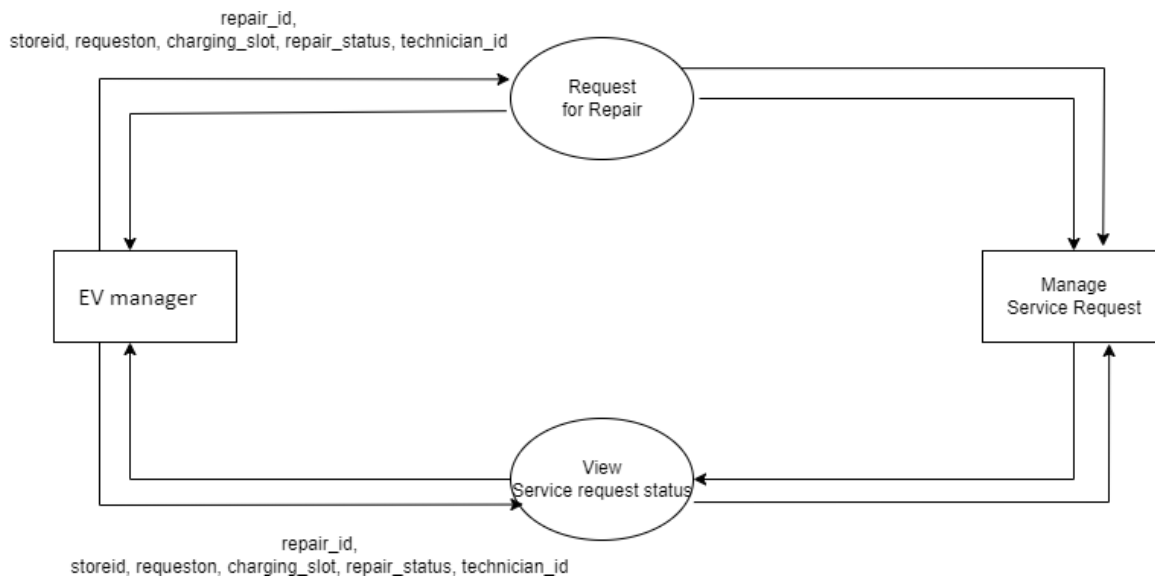
3.6.13 DFD level 1(EV manager)

3.6.14 DFD level 2(Slot Tracker):



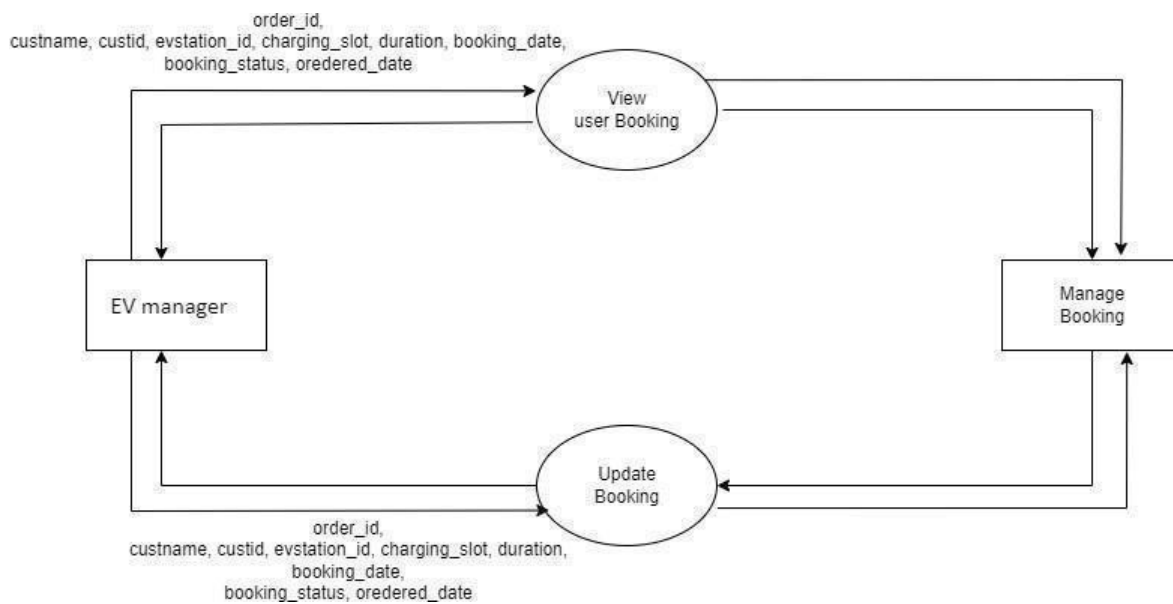
3.6 .14 DFD level 2(Slot tracker)

3.6.15 DFD level 2(Service Request):



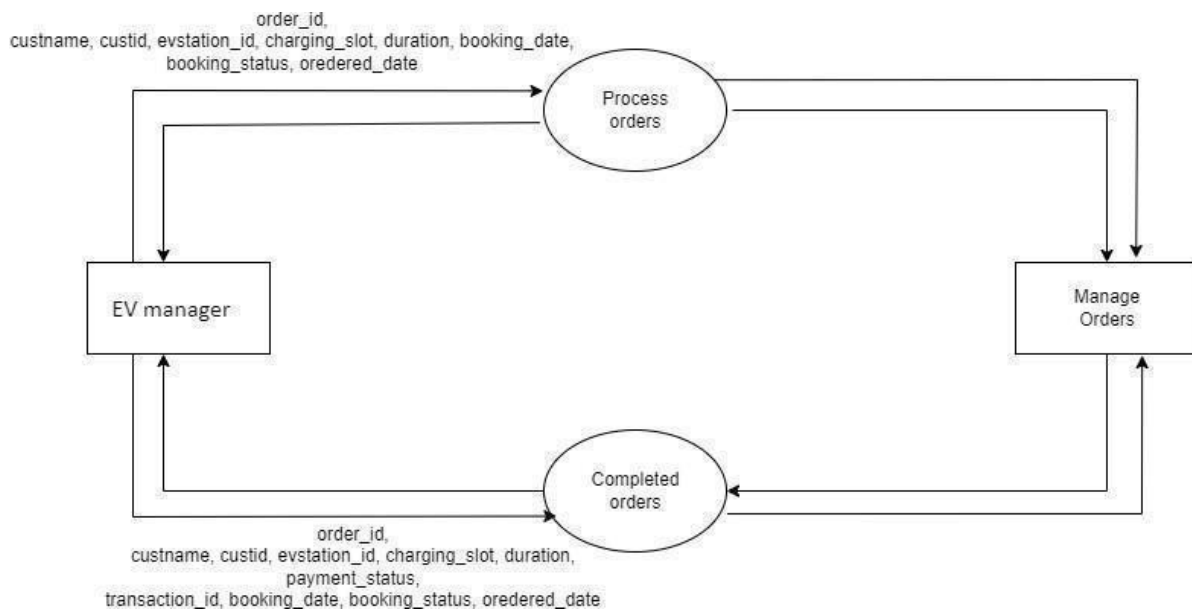
3.6.15 DFD level 2 (Service request)

3.6.16 DFD level 2 (Booking):



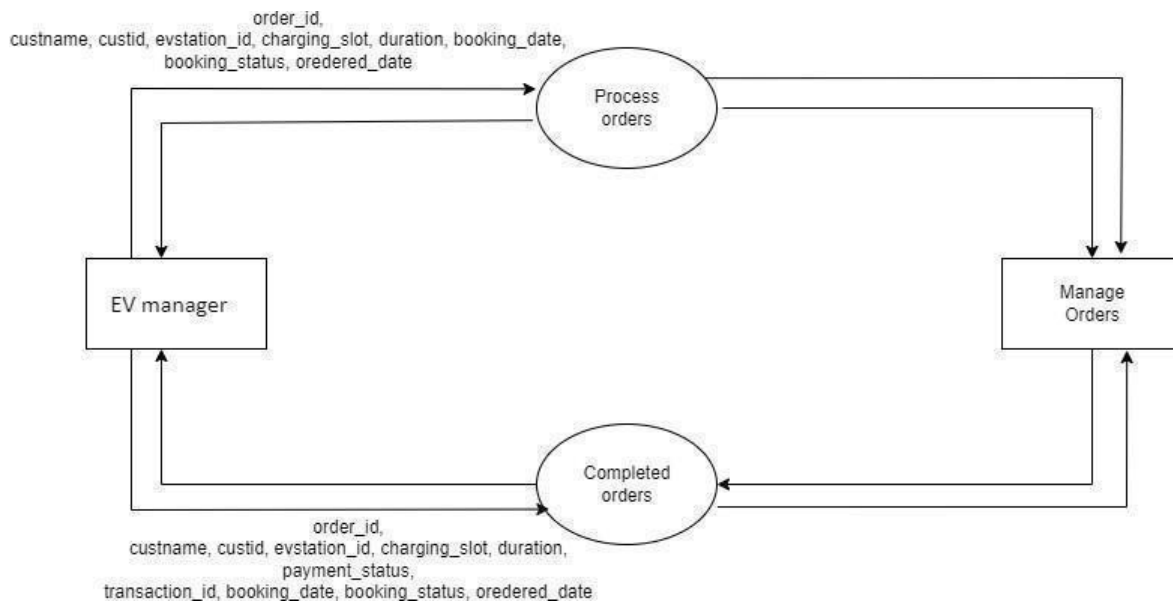
3.6.16 DFD level 2 (Booking)

3.6.17 DFD level 2(Orders):



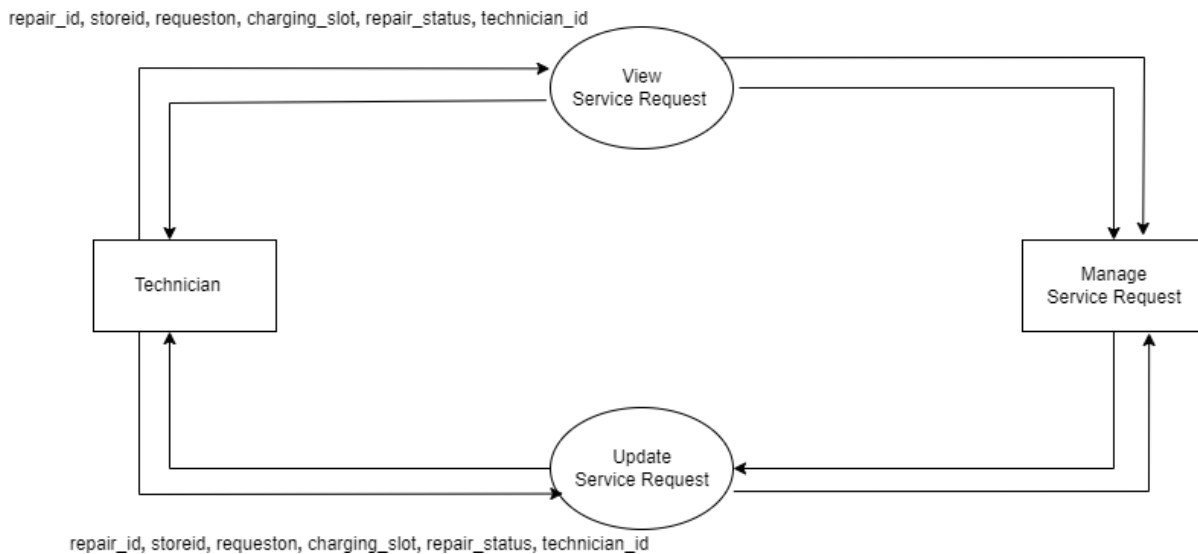
3.6.17 DFD level 2 (Orders)

3.6 .18 DFD level 1(Technician):



3.6.18 DFD level 1 (Technician)

3.6.19 DFD level 2(Service Request):



3.6.19 DFD level 2 (Service request)

3.7 Entity Relationship Diagram

The ER Diagram includes key parts such as Data objects, Attributes, Relationships, and various type indicators. Its main purpose is to display Data objects and the ways they are connected to each other. By understanding these connections, we can better grasp how different pieces of data interact within a system.

3.7.1 Data Objects

A data object is something that shows combined information, which software needs to understand. This combined information includes many different features or characteristics.

3.7.2 Attributes

Attributes are important features that describe data objects, and they come with three main traits. First, they can give a name to a specific instance of a data object. Second, they can provide a description of the instance. Third, attributes can connect or point to another instance located in a different table. It's also necessary to have at least one attribute act as an identifier. This identifier becomes what we call a "key," and it's very useful when we need to find or locate a particular instance of the data object.

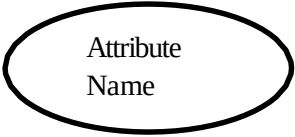
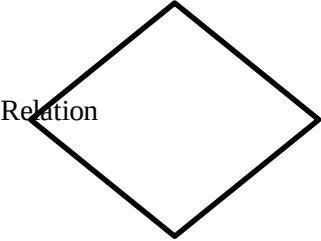
3.7.3 Relationships

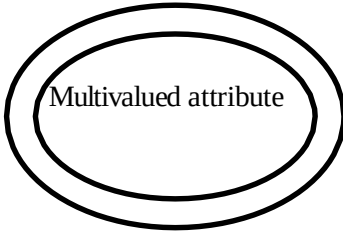
Indicate the manner in which data objects are “connected” to one another

3.7.4 Cardinality

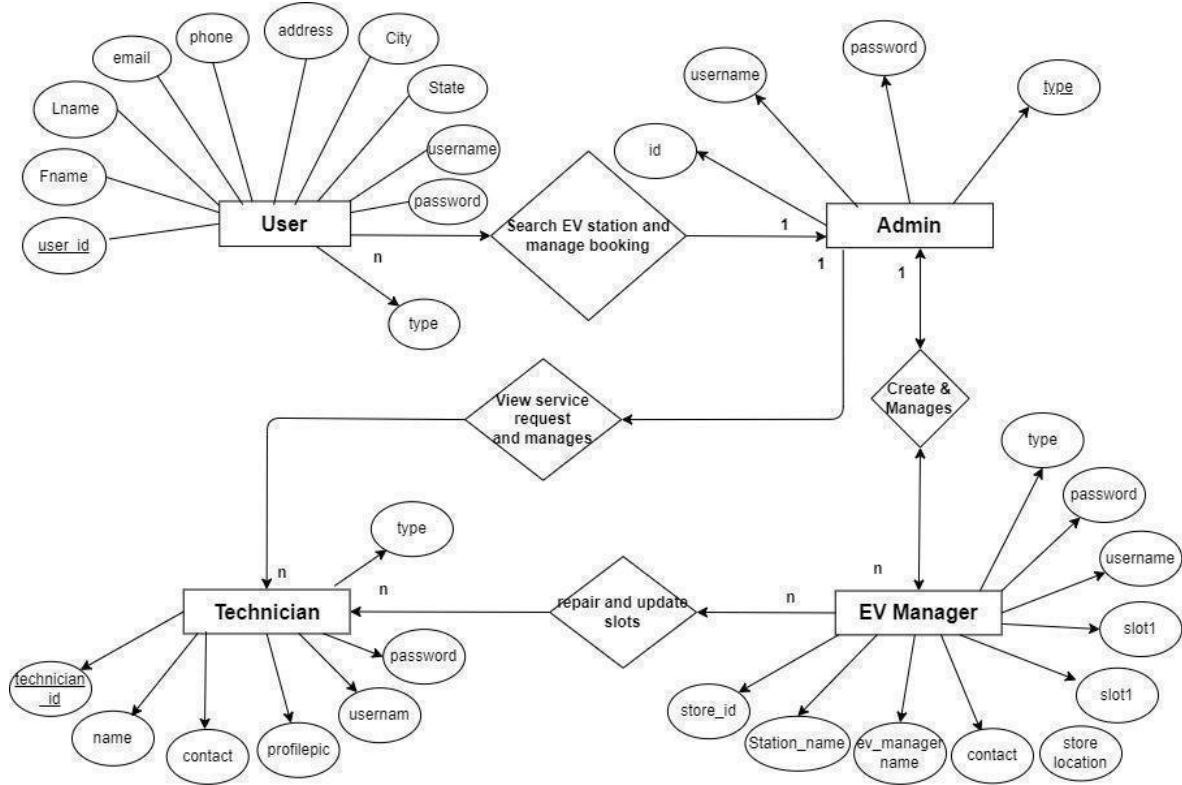
The data model needs to be able to show how often objects appear in specific relationships. Tillman explains cardinality as a key part of this. Cardinality describes the number of times one object can link to another object. It is usually shown as "one" or "many." By looking at all the possible combinations of "one" and "many," two objects can relate in various ways.

The Symbols are shown in below table:

Name	Notation	Description
Entity		It may be an object with the physical existence or conceptual existence. It is represented by a Rectangle.
Attribute		The properties of the entity can be attribute. It is represented by an Ellipse.
Relationship		Whenever an attribute of one entity refers to another entity, some relationship exists. It is represented by a Diamond.
Link		Lines link attributes to entity sets and entity sets to relation.
Derived attribute		Dashed ellipses denote derived attributes.

Key Attribute		An entitytype usually has an attribute whose values are distinct for each individual entry in the entity set. It is represented by a Underlined word in ellipse.
Multivalued Attribute		Attributes that have different numbers of values for a particular attribute. It is represented by a Double ellipse represents multi-valued attributes.
Cardinality Ratio	1) 1:1 2) 1:M 3) M:1 4) M:M	It specifies the maximum number of relationships instances that an entity can participate in. There are four cardinality ratios.

3.8 ER diagram:



3.8 ER diagram

4. DATABASE DESIGN

4. DATABASE DESIGN

4.1 Introduction:

- Database: A database is a group of related data. It can be large or small and vary in complexity. Databases are useful for storing and retrieving data easily. The main goal of a database is to provide the information needed by a system according to its specific requirements.
- Database Design: Before creating a database, we must design it to meet the needs of the users who will use it in a particular information system. The design outlines what data is necessary and how it is organized within the database. We use MySQL to physically build and implement the database.

4.2 Scope

- A database system is considered self-describing since it contains both the actual data and metadata. Metadata explains the data and details how the tables within the database are connected. This additional information can be accessed by the DBMS software or by those using the database whenever it's necessary.

4.3 Database structure

4.3.1 Structure of Table "Area":

Description: The **Area table** stores information about different areas, including their unique ID, name, and the date they were uploaded.

Column Name	Data Type	Size	Constraints	Description
id	int	11	Primary Key	Table name ID
areaname	varchar	100	Not Null	Area name
uploaded_on	date	-	Not Null	Upload date

4.3.2 Structure of Table “BookSlots”:

Description: The **Bookslot table** stores EV charging booking details, including order ID, customer information, charging slot, duration, payment status, and booking dates

Column Name	Data Type	Size	Constraints	Description
order_id	int	11	Primary Key	Table name ID
custname	char	50	Not Null	Customer name
custid	int	11	Not Null	Customer ID
evstation_id	int	11	Not Null	EV charging station ID
charging_slot	varchar	50	Not Null	Charging slot details
duration	varchar	50	Not Null	Charging duration
payment_status	char	30	Not Null	Payment status
transaction_id	varchar	20	Not Null	Payment transaction ID
booking_date	date	-	Not Null	Date of booking
booking_status	char	50	Not Null	Status of the booking
oredered_date	date	-	Not Null	Date when ordered

4.3.3 Structure of Table “EV StationTable”:

Description: The **EV Station Table** stores details of EV charging stations, including station name, manager, location, contact information, slot availability, and operational status.

Column Name	Data Type	Size	Constraints	Description
storeid	int	11	Primary Key	Table name ID
stationname	varchar	30	Not Null	EV station name
ev_managename	varchar	30	Not Null	EV station manager name
contact	bigint	20	Not Null	Contact number
	varchar	50	Not Null	Location ofthe EV station
storeaddress	varchar	50	Not Null	Address ofthe EV station
slot1	char	20	Not Null	Slot 1 availability/status
slot2	char	20	Not Null	Slot 2 availability/status
station_status	char	20	Not Null	Status of the EV station
username	varchar	30	Not Null	Associated username

4.3.4 structure of Table “Login”:

Description: The **LoginTable** stores user authentication details, including username, password, and role-based access control.

Column Name	Data Type	Size	Constraints	Description
id	int	11	Primary Key	Table name ID
username	varchar	30	Not Null	User login name
password	varchar	30	Not Null	User password
role	varchar	20	Not Null	User role type

4.3.5 Structure of Table “Slot Tracker”:

Description: The **Slot Tracker Table** records EV charging slot details, including slot ID, station ID, charging duration, and booking date.

Column Name	Data Type	Size	Constraints	Description
slot_id	int	11	Primary Key	Table name ID
storeid	int	11	Not Null	EV station ID
charging_slot	varchar	30	Not Null	Charging slot identifier
duration	varchar	30	Not Null	Charging duration
booking_date	varchar	30	Not Null	Date of slot booking

4.3.6 Structure of Table “Technician”:

Description: The **Technician Table** stores technician details, including their name, contact information, profile picture, and associated username.

Column Name	Data Type	Size	Constraints	Description
technician_id	int	11	Primary Key	Table name ID
name	varchar	30	Not Null	Technician name
contact	bigint	20	Not Null	Technician contact
profilepic	char	50	Not Null	Profile picture path
username	varchar	30	Not Null	Associated username

4.3.7 Structure of Table “TechnicianRequest”:

Description: The **Technician Request Table** tracks repair requests for EV charging stations, including station ID, charging slot, request date, repair status, and assigned technician.

Column Name	Data Type	Size	Constraints	Description
repair_id	int	11	Primary Key	Table name ID
storeid	int	11	Not Null	EV station ID
requeston	date	-	Not Null	Date of repair request
charging_slot	char	30	Not Null	Charging slot requiring repair
repair_status	varchar	30	Not Null	Current repair status
technician_id	int	11	Not Null	Assigned technician ID

4.3.8 Structure of Table “User Register”:

ColumnName	Data Type	Size	Constraints	Description
user_id	int	11	Primary Key	Table name ID
fname	char	20	Not Null	First name
lname	char	20	Not Null	Last name
email	varchar	30	Not Null, Unique	User email ID
phone	bigint	20	Not Null, Unique	Contact number
address	varchar	100	Not Null	User address
city	char	30	Not Null	City name
state	char	30	Not Null	State name
username	varchar	50	Not Null, Unique	User login name
password	varchar	50	Not Null	User password
role	varchar	15	Not Null	User roletype

5. DETAILED DESIGN

5. DETAILED DESIGN

5.1 Introduction

This document is a design report for the “Chargify EV.” It focuses on guiding you through system design and implementation. Detailed design is the second step in the process. At this stage, we describe how each part of the system interacts with the others and establish the internal logic for each module. This is why it's sometimes called logic design. Detailed design takes the initial system and database design and expands on them. It provides a thorough description of how processes function and what data structures are used, making the design ready for coding.

5.2 Applicable Documents

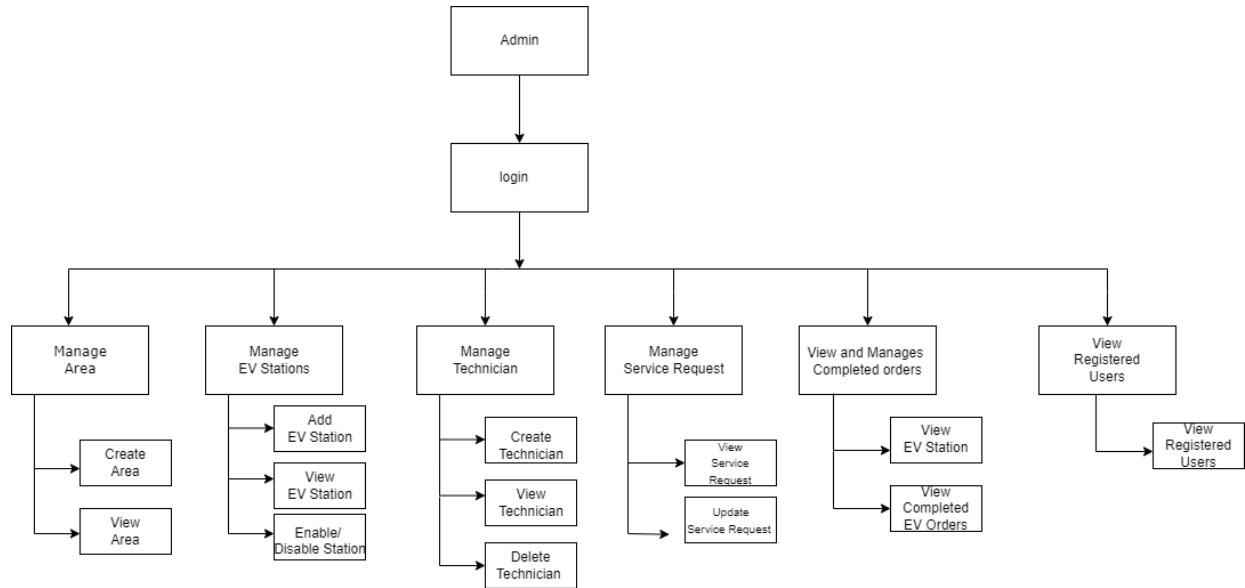
- Synopsis document for “Chargify EV”
- Software Requirement Specification (SRS) for “Chargify EV”.
- Database Design document for “Chargify EV”

5.3 Structure of Software Package

The functional components for:

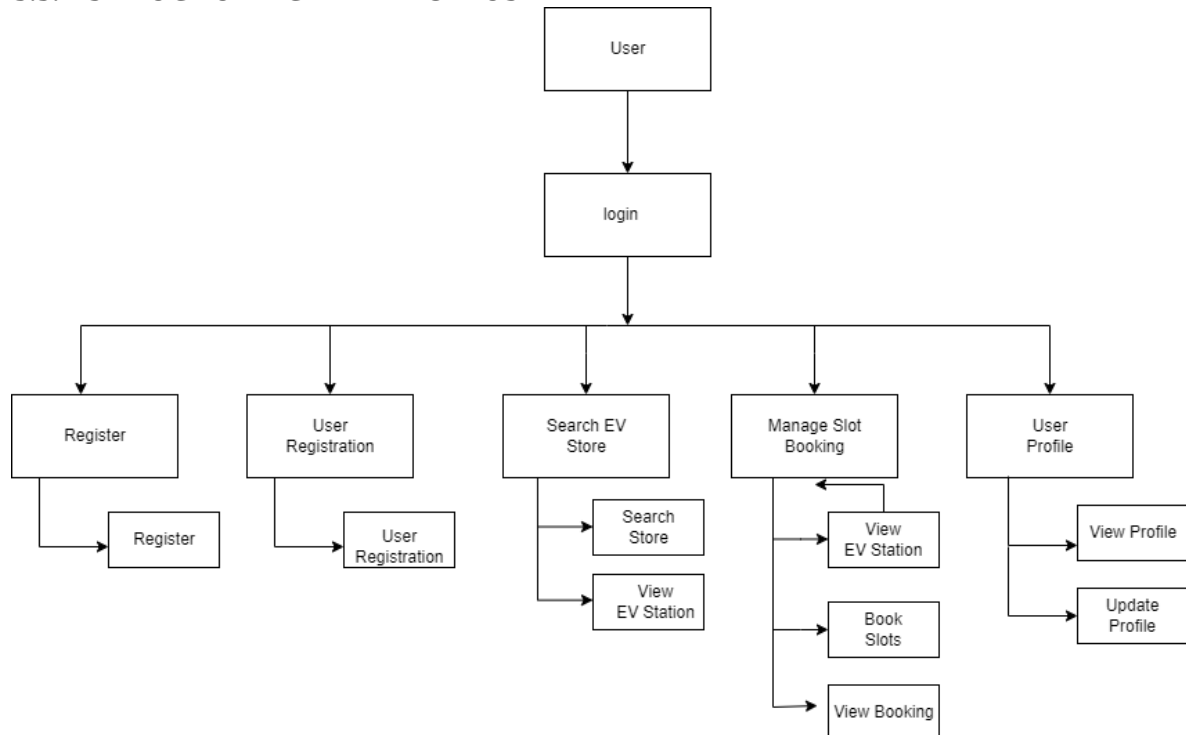
- Admin
- User
- EVManager
- Technician

5.3.1 STRUCTURE CHART FOR ADMIN



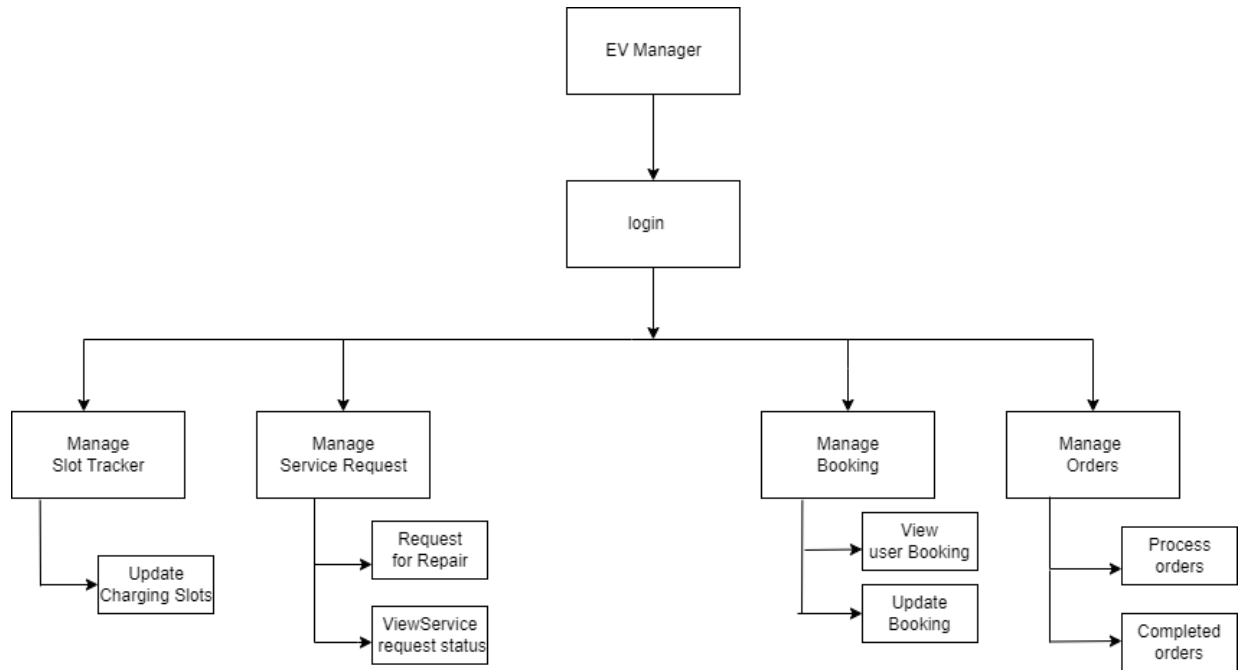
5.3.1 Structure Chart for Admin

5.3.2 STRUCTURE CHART FOR USER



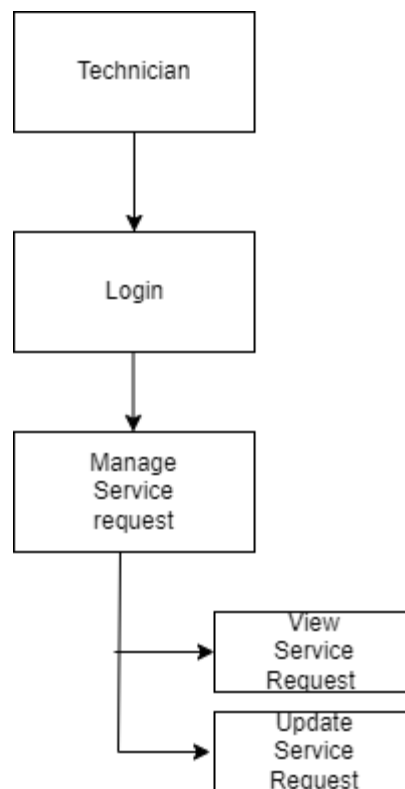
5.3.2 Structure Chart for User

5.3.3 STRUCTURE CHART FOR EV MANAGER



5.3.3 Structure Chart for EV Manager

5.3.4 STRUCTURE CHART FOR TECHNICIAN



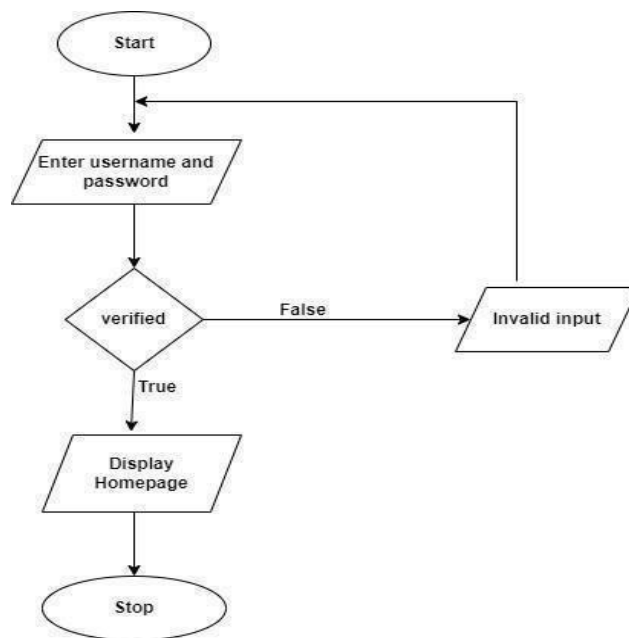
5.3.4 Structure Chart for Technician

5.4 Modular Decomposition of Elements

5.4.1 ADMIN

a) Login

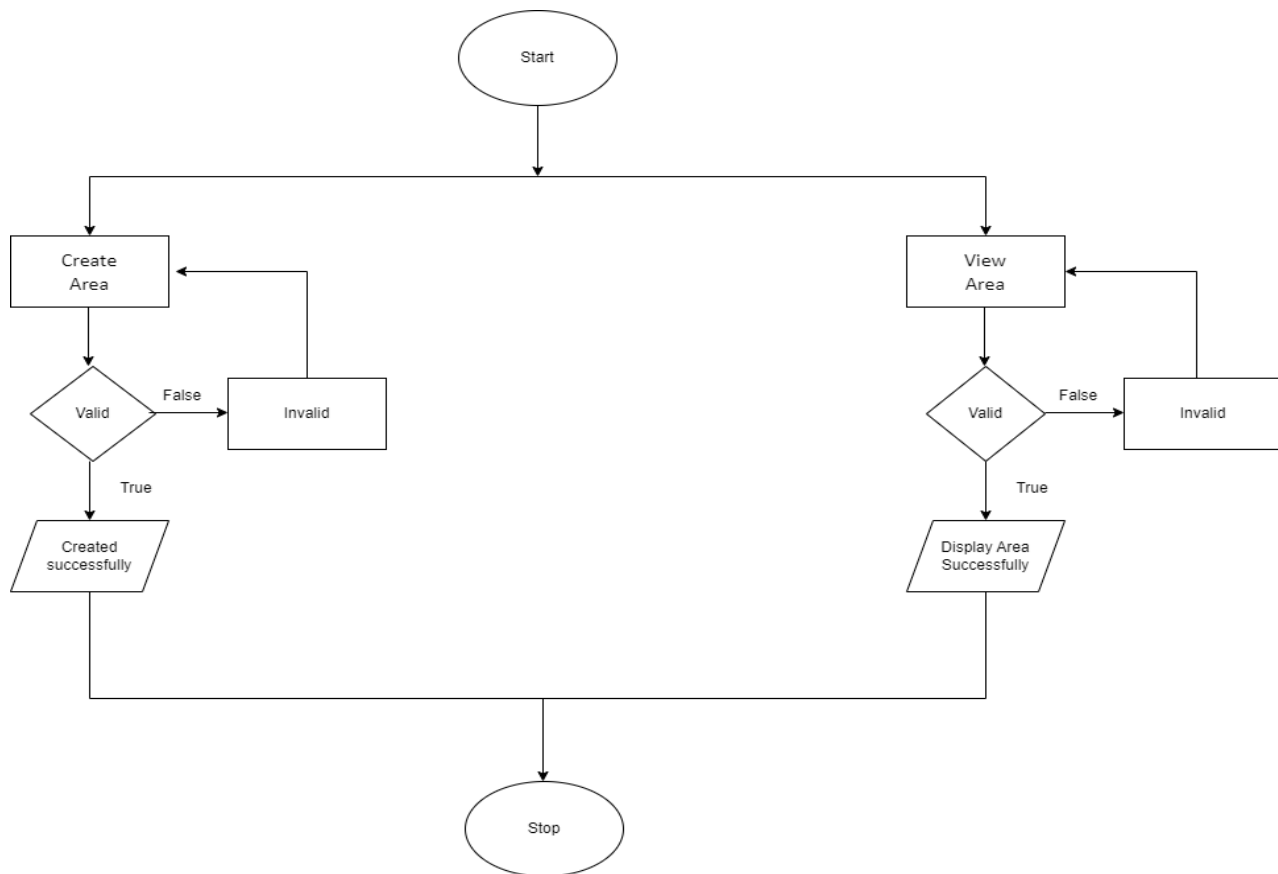
- **Input:** Id, name, username, password, type
- **Procedural details:**



Login

b) Manage Area

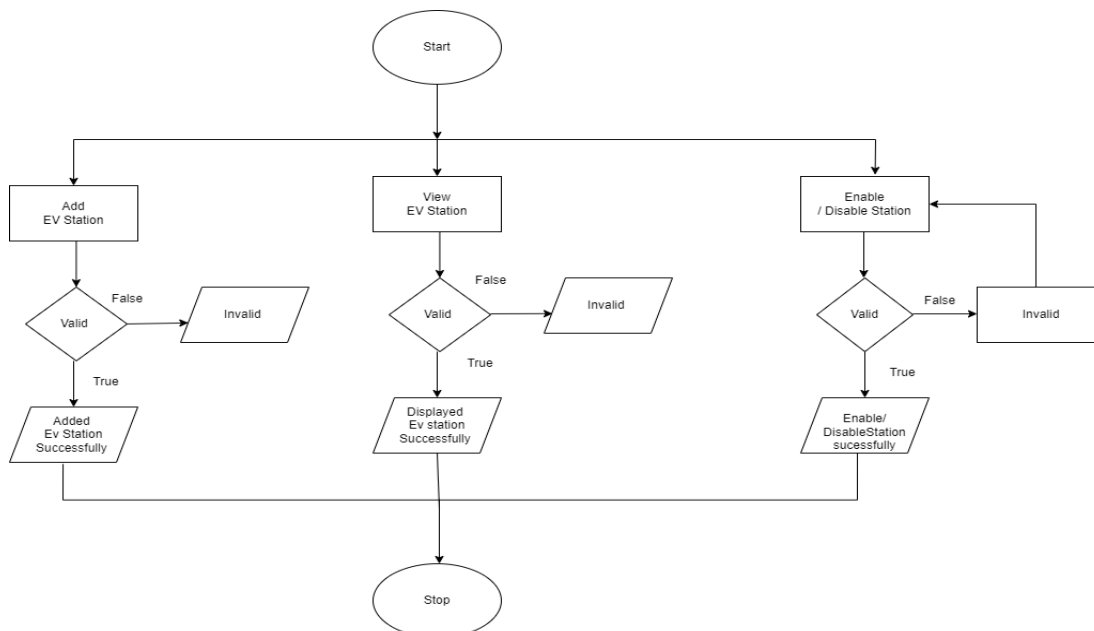
- **Input:** : Id, areaname, uploaded_on
- **Procedural details:**



Manage area

C) Manage EV Station

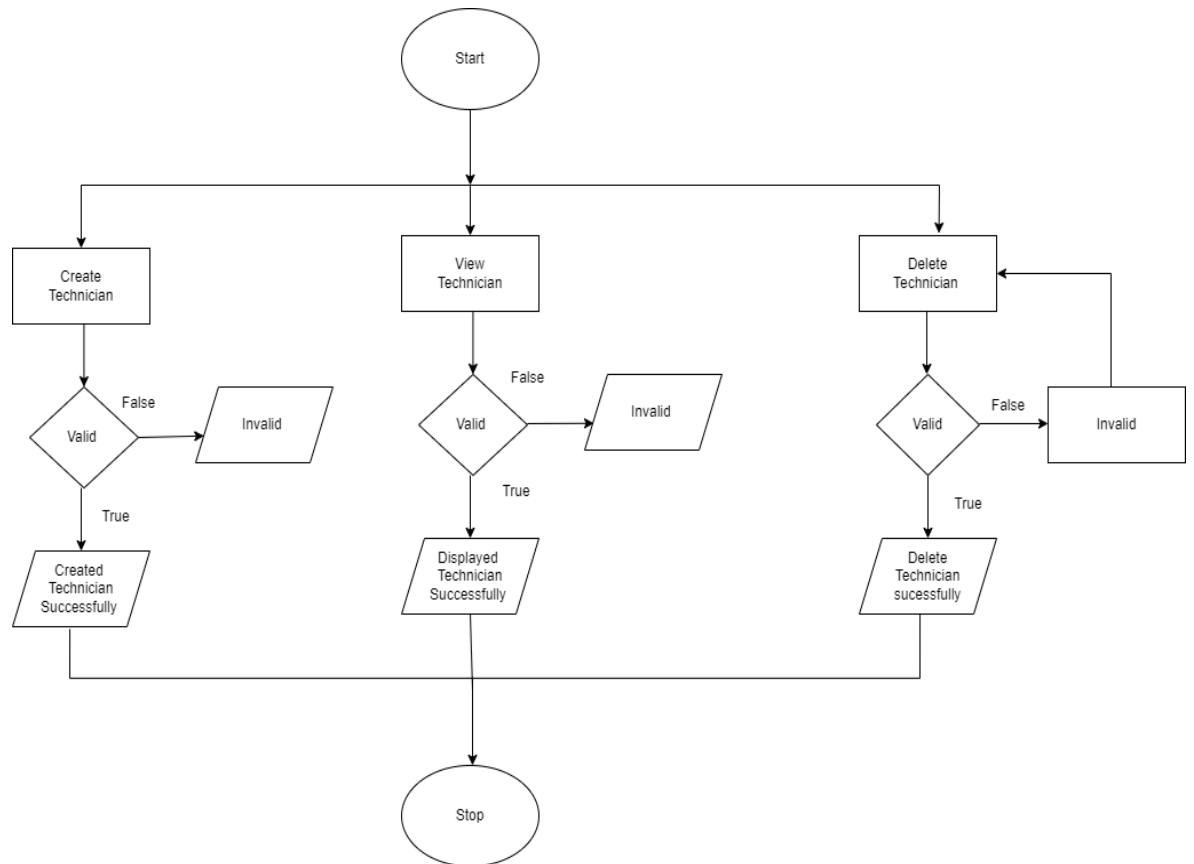
- **Input:** Store_id, stationname, ev_managename, contact, storelocation, storeaddress, slot1, slot2, station_status, username
- **Procedural details:**



Manage EV Station

d) Manage Technician

- **Input:** technician_id, name, contact, profilepic
- **Procedural details:**

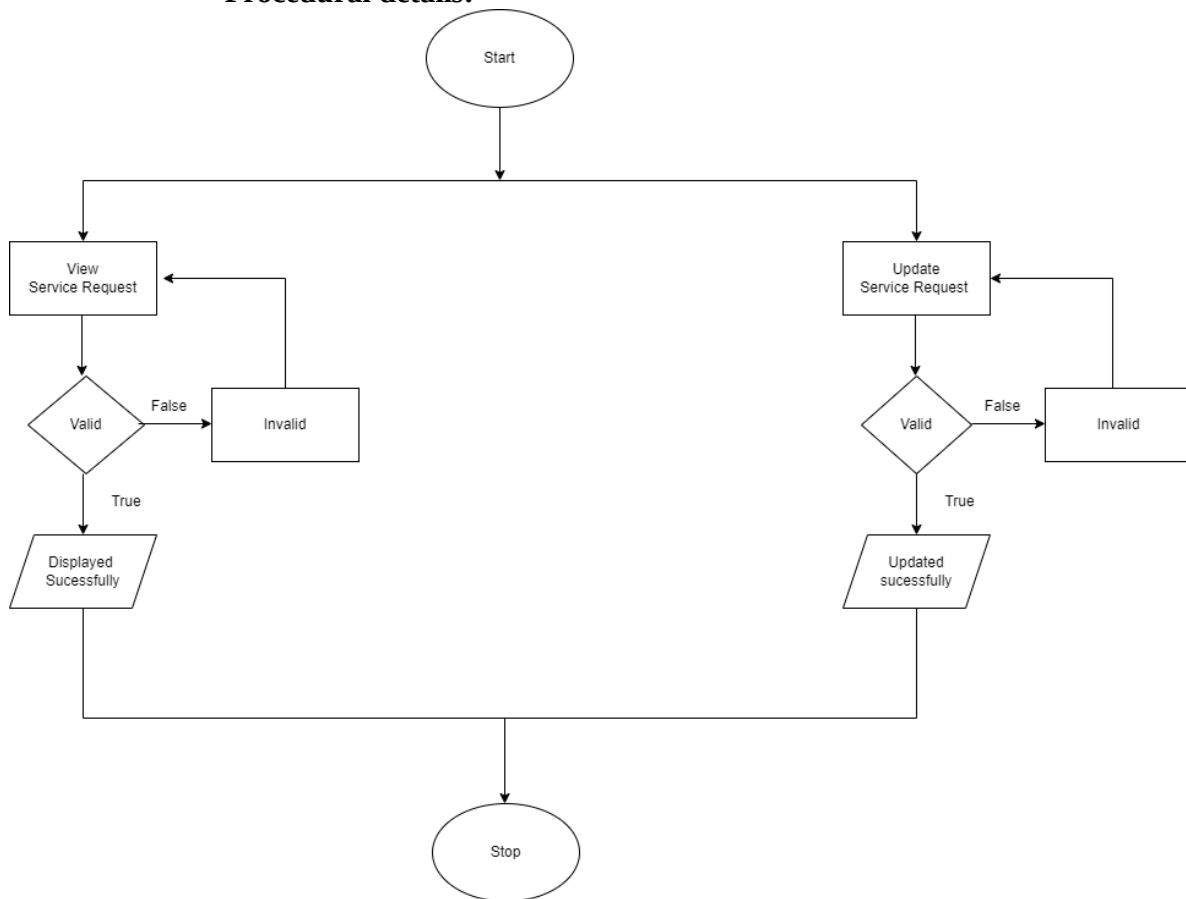


Manage technician

e) Manage Service Request

- **Input:** repair_id, storeid, requeston, charging_slot, repair_status

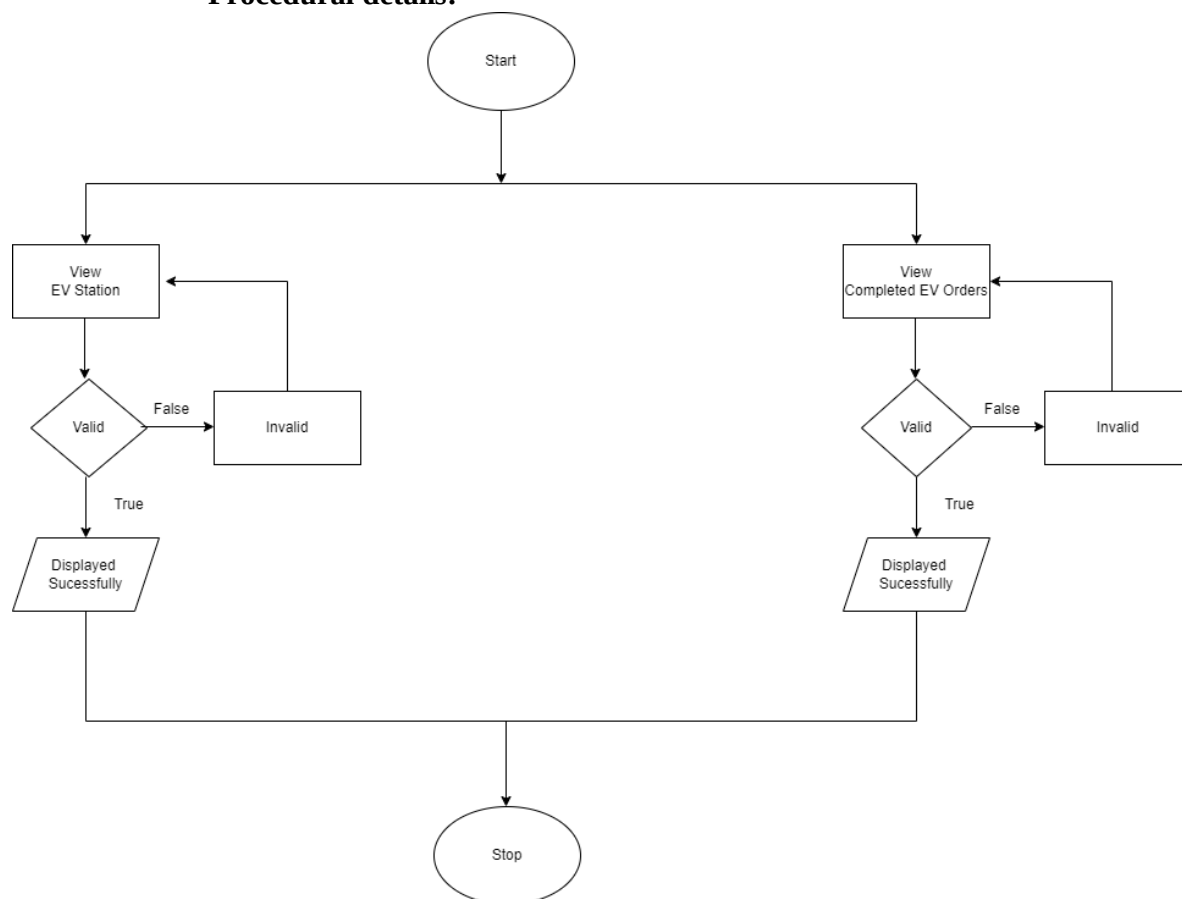
Procedural details:



Manage service request

f) View and Manages Completed orders

- **Input:** Storied, stationname, ev_managername, contact, storelocation, storeaddress
- **Procedural details:**

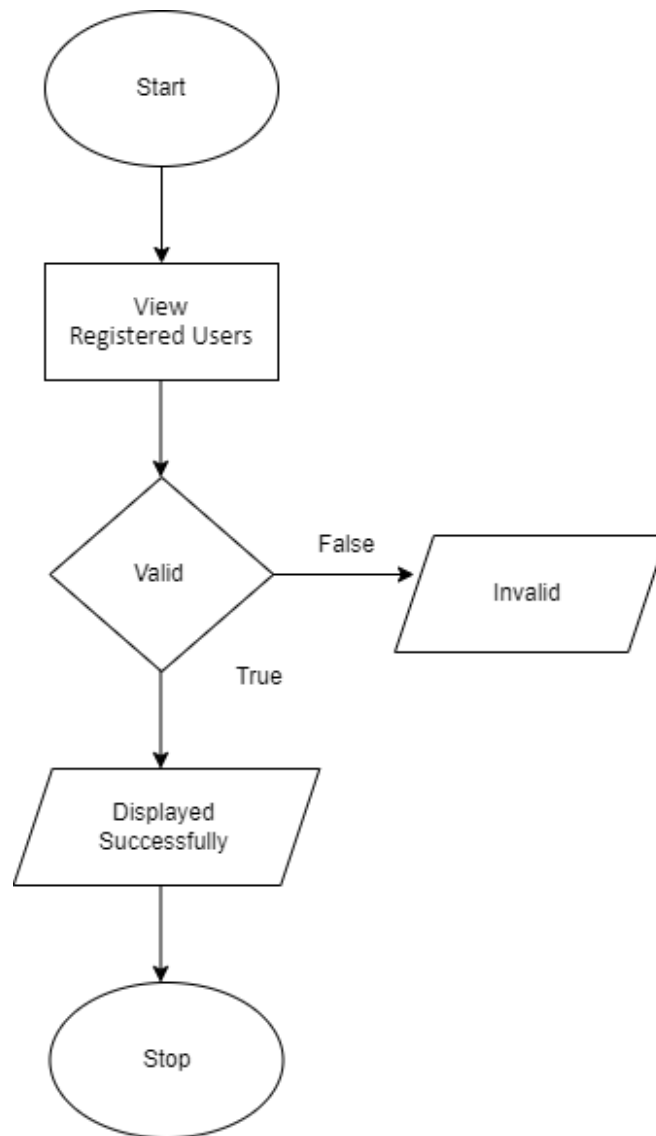


View and manages completed orders

g) View Registered Users

a. **Input:** user_id, fname, lname, email, phone, address, city, state

b. **Procedural details:**

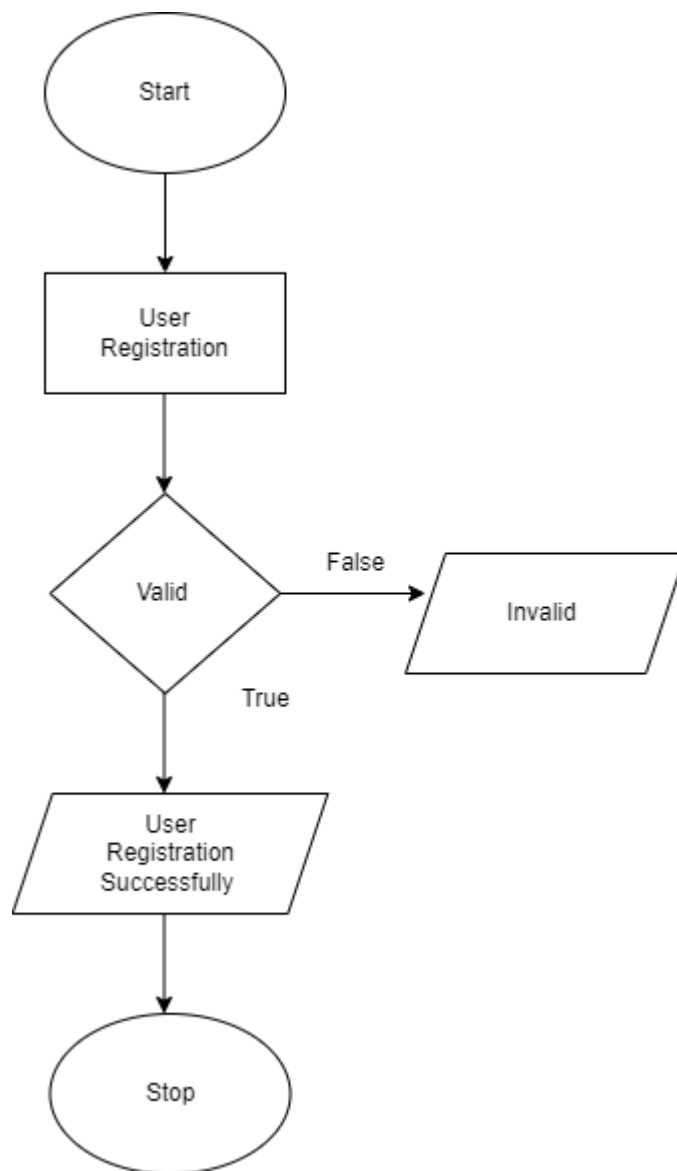


View registered users

5.4.2 User

a) User Registration

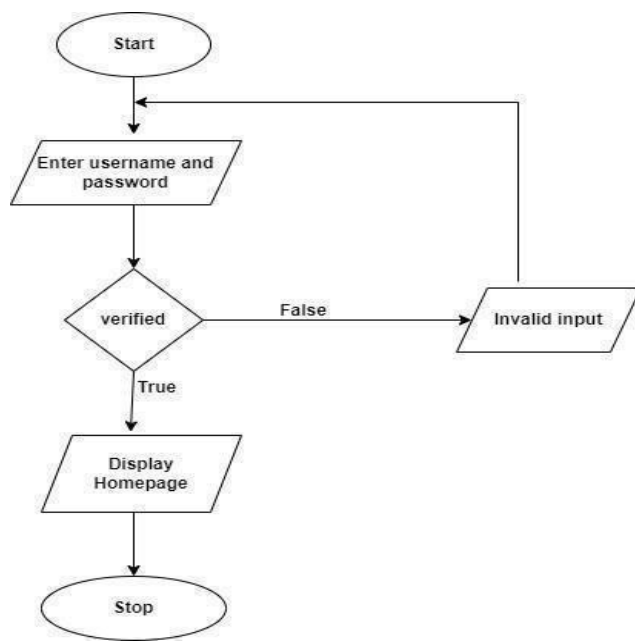
- **Input:** user_id, Fname, lname, email, phone, address, city, state, username, password, role
- **Procedural details:**



User

b) Login

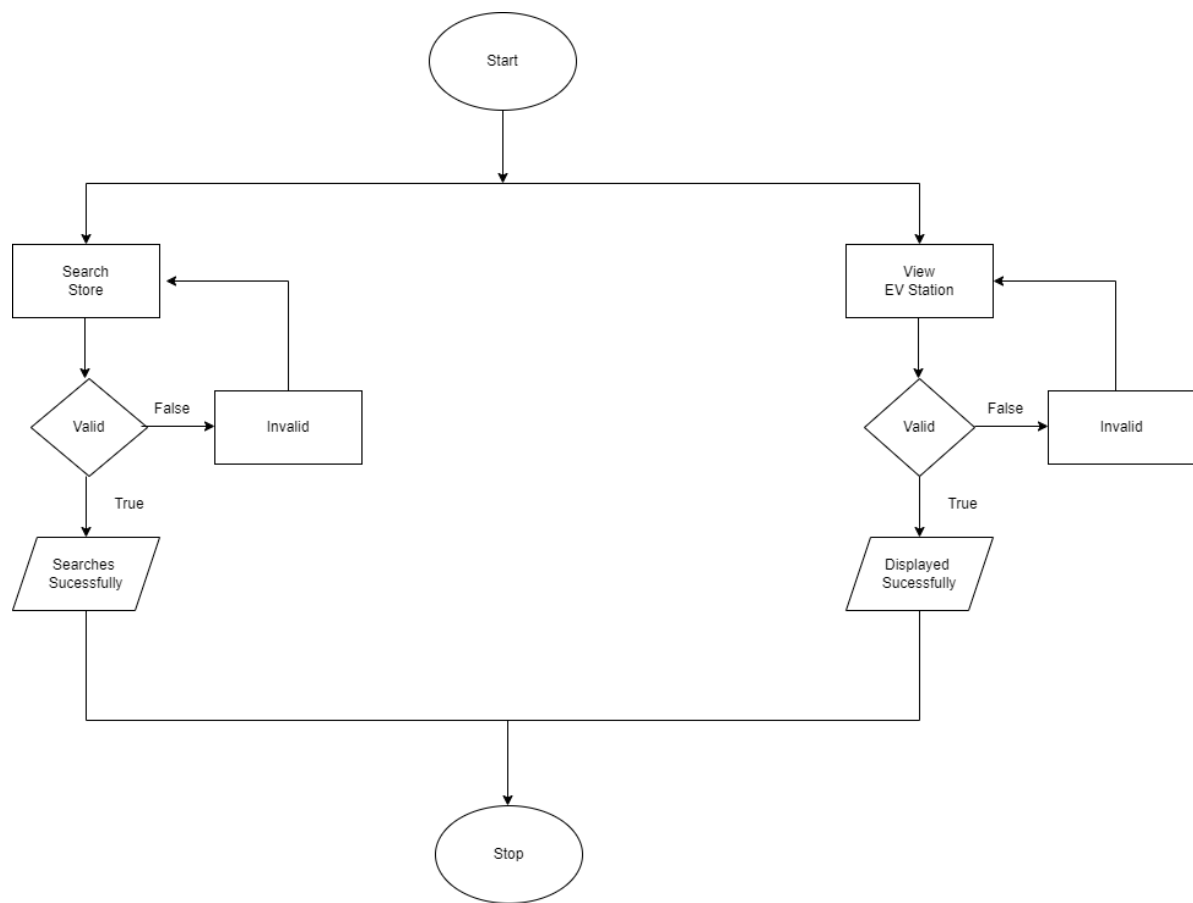
- **Input:** Id, name, username, password, type
- **Procedural details:**



Login

c) Search EV Store

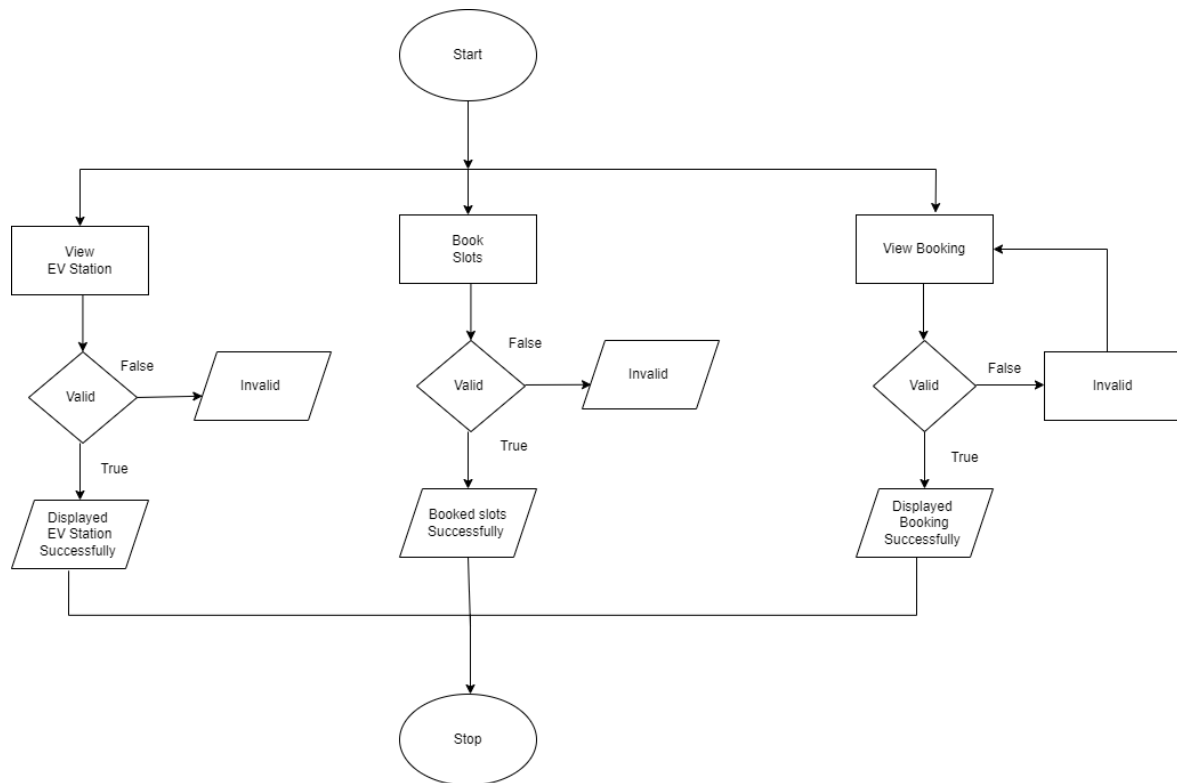
- **Input:** Store_id, stationname, ev_managername, contact, storelocation, storeaddress, slot1, slot2, station_status
- **Procedural details:**



Search EV Store

d) Manage Slot Booking

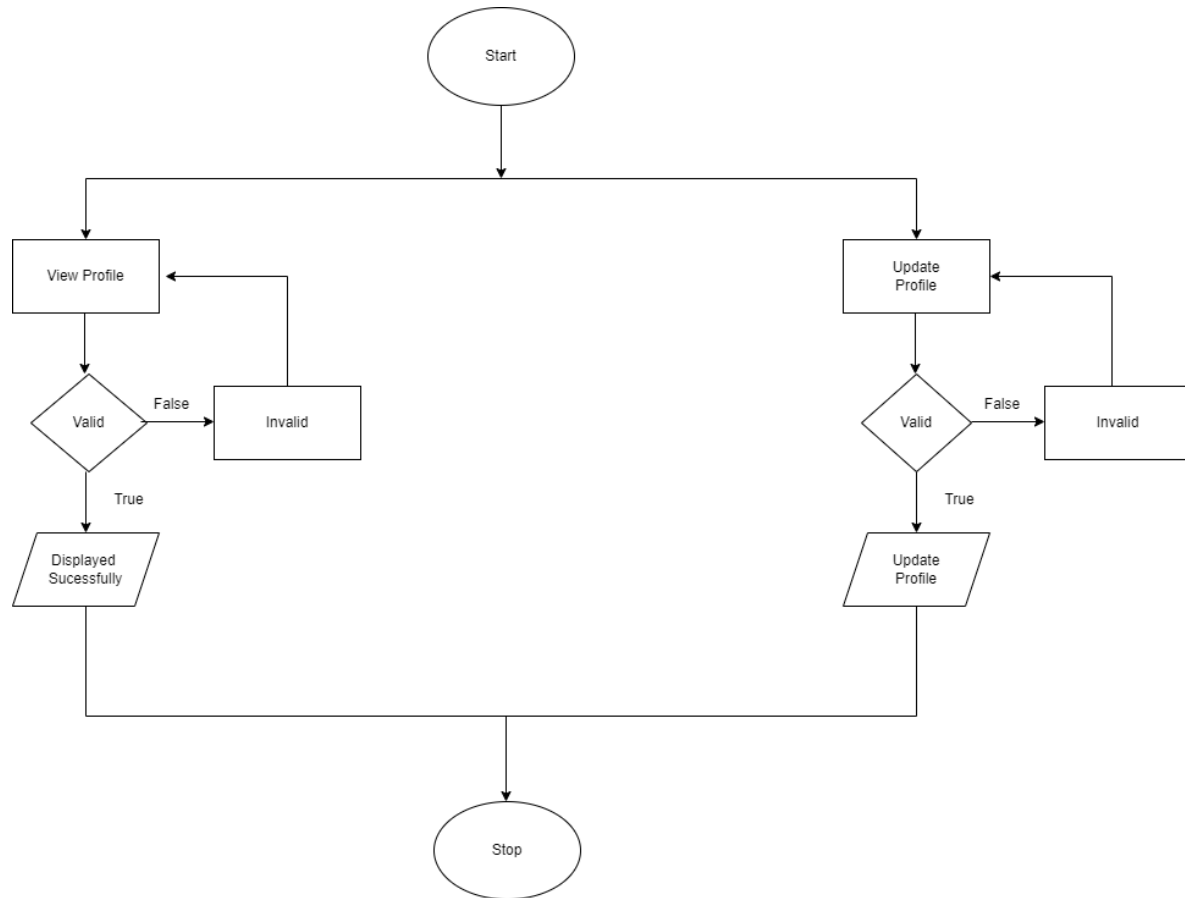
- **Input** : Storied, stationname, evView EV Station
_managername, contact, storelocation, storeaddress, slot1, slot2,
station_status
- **Procedural details:**



Manage slot booking

e) User Profile

- **Input** : user_id, Fname, lname, email, phone, address, city, state
- **Procedural details:**

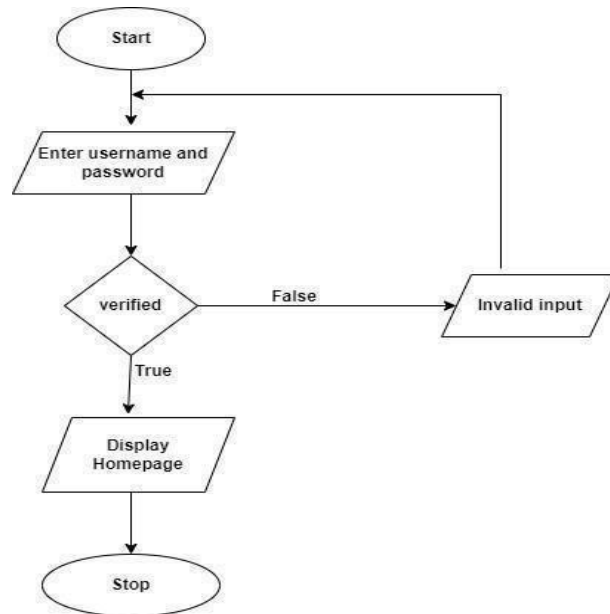


User profile

5.4.3 Ev Manager

a) Login

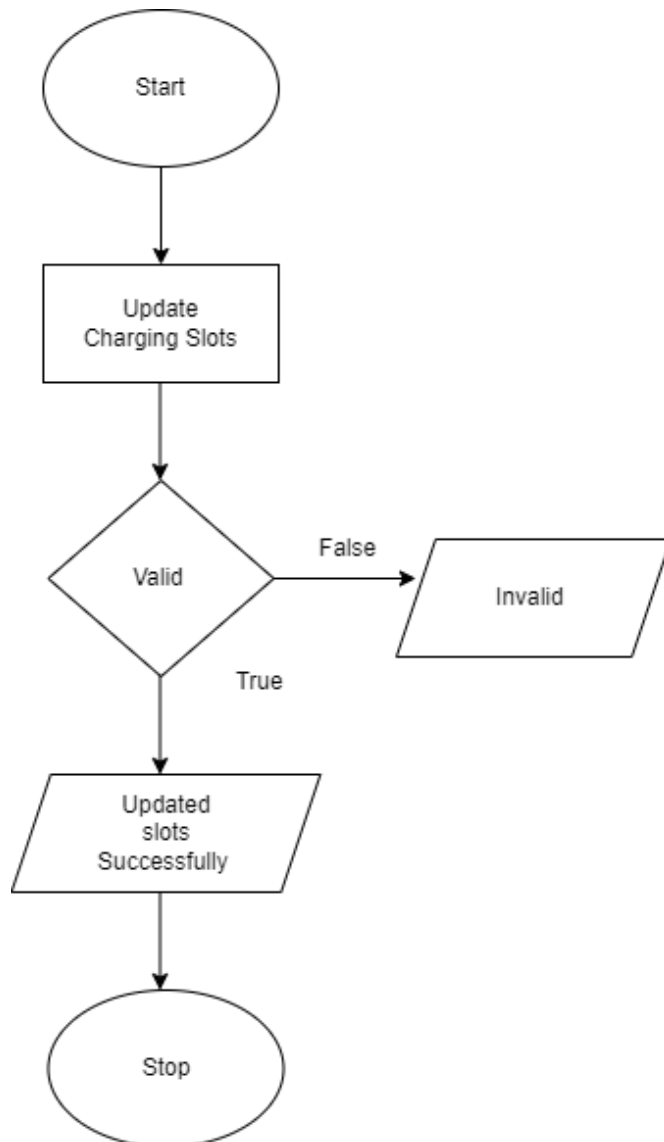
- **Input:** Id, name, username, password, type
- **Procedural details:**



Ev manager

b) Manage Slot Tracker

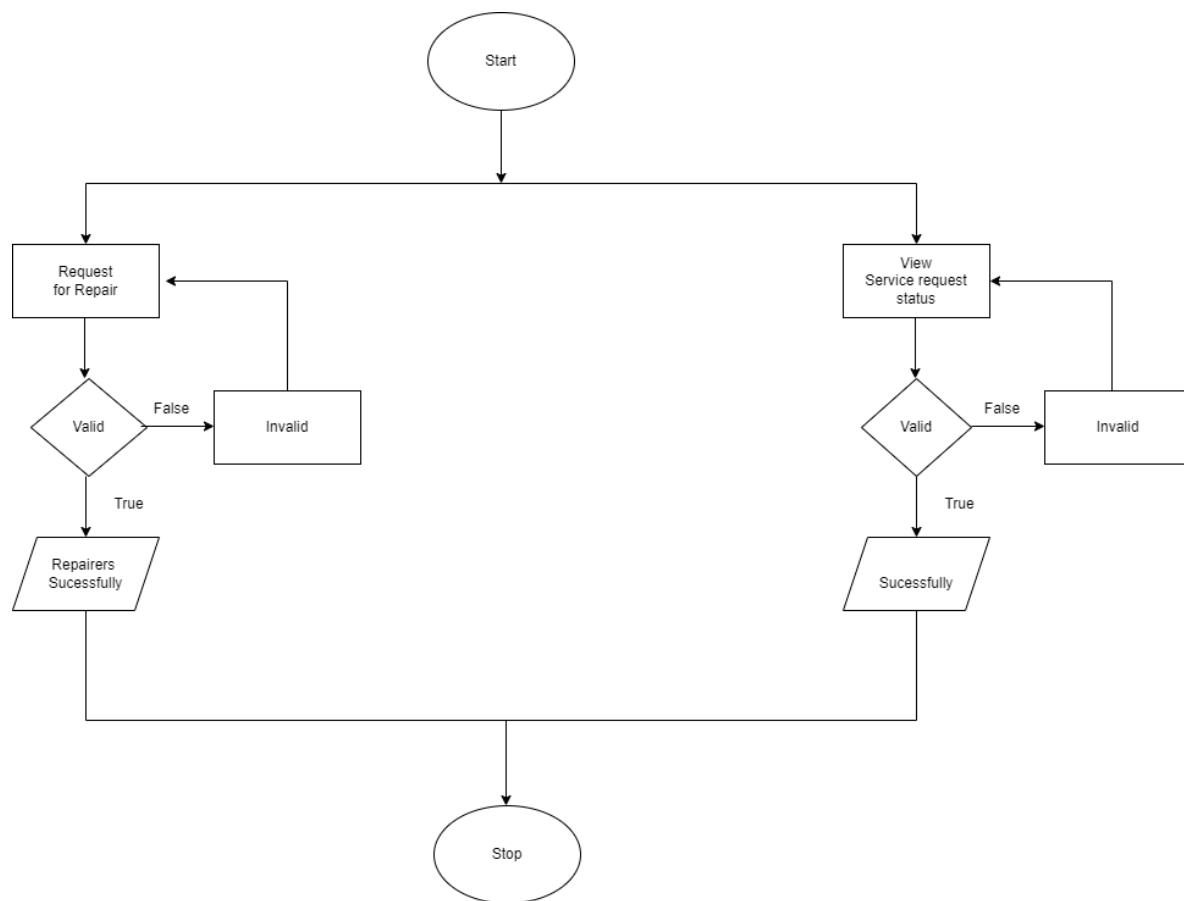
- **Input :** slot_id, storeid, charging_slot, duration, booking_date
- **Procedural details:**



Manage slot tracker

c) Manage Service Request

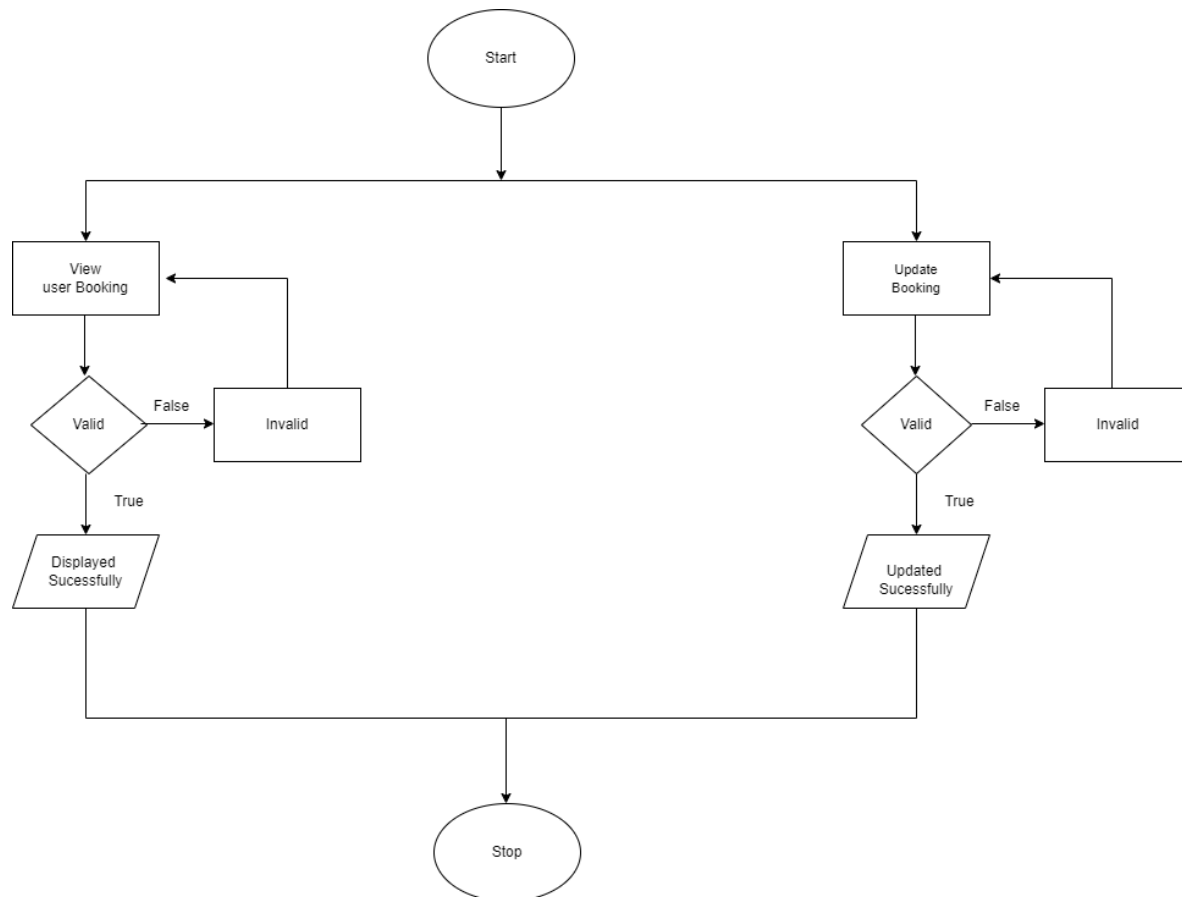
- **Input** : repair_id, storeid, requeston, charging_slot, repair_status, technician_id
- **Procedural details:**



Manage service request

d) Manage Booking

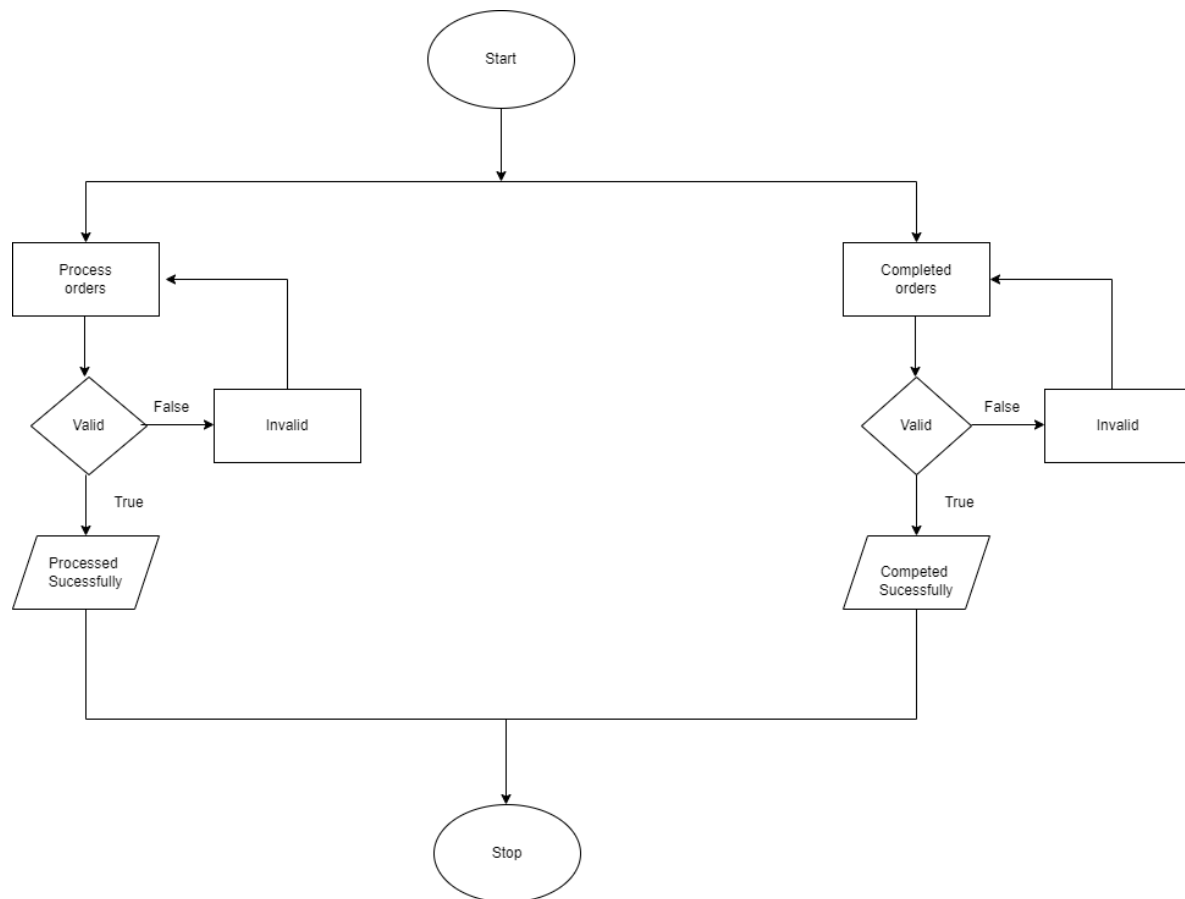
- **Input** : order_id, custname, custid, evstation_id, charging_slot, duration, booking_date, booking_status, ordered_date
- **Procedural details:**



Manage booking

e) Manage Orders

- **Input** : order_id, custname, custid, evstation_id, charging_slot, duration, payment_status, transaction_id, booking_date, booking_status, ordered_date
- **Procedural details:**

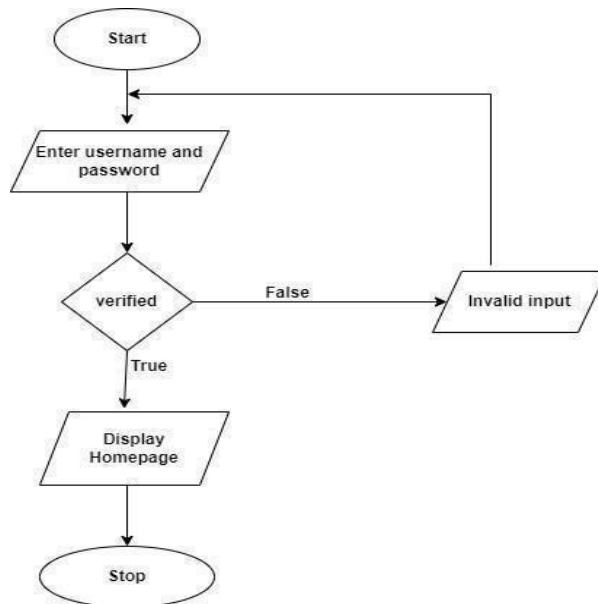


Manage orders

5.4.4 Technician

a) Login

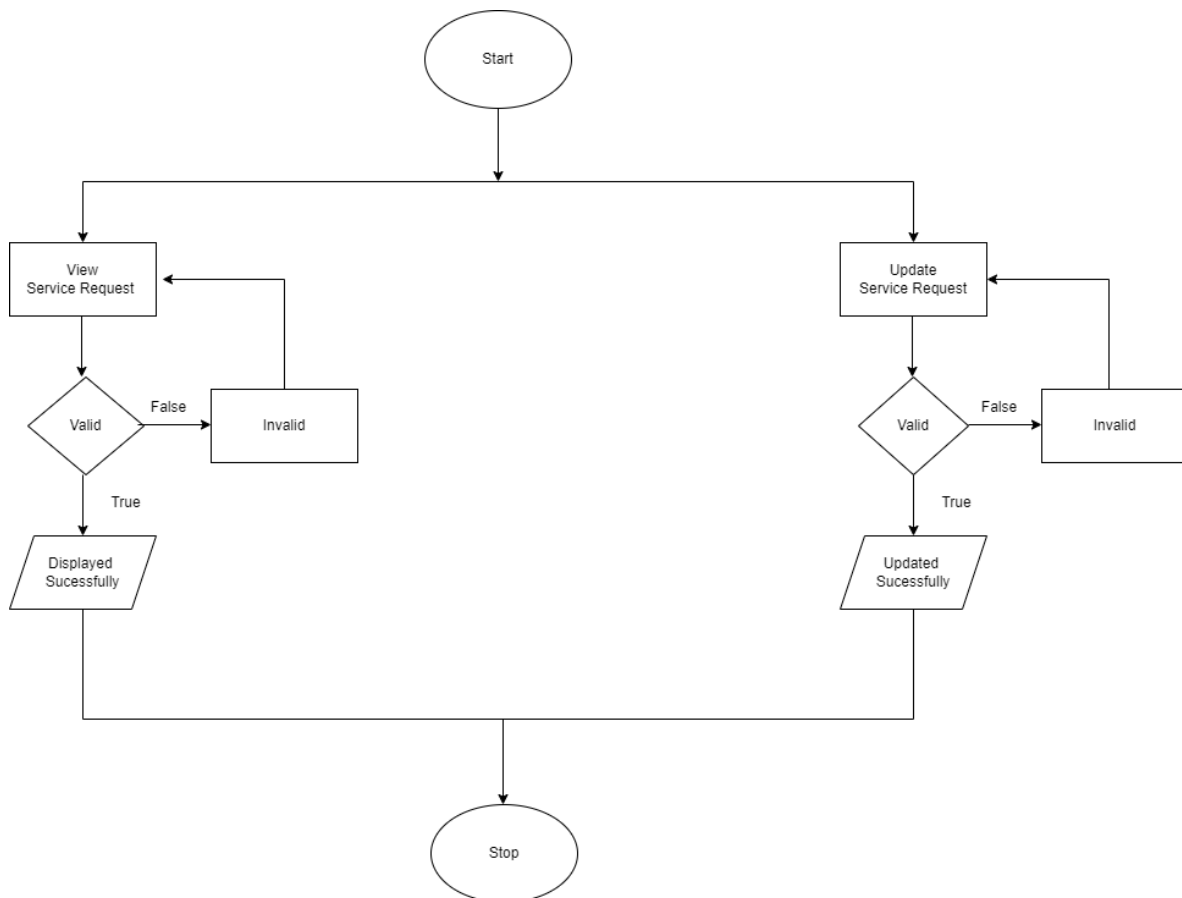
- **Input:** Id, name, username, password, type
- **Procedural details:**



Technician

b) Manage Service Request

- **Input:** repair_id, storeid, requeston, charging_slot, repair_status, technician_id
- **Procedural details:**



Manage Service Request

6. CODING

6. CODING

Login page:

```
def login_req(request):
    if request.method=="POST":
        usr_name=request.POST['username']
        userpassword=request.POST['password']
        role=request.POST['role']

    if role=="user":
        check_user=login.objects.filter(Q(username=usr_name) & Q(password=userpassword)).first()
        if check_user:
            #print(check_user.id)

            request.session['user']=usr_name
            request.session['userid']=check_user.id
            messages.success(request,"Login Success," , usr_name)
            return redirect ('userhome')
        else:
            messages.warning(request,"Please enter valid Username, User not Registered")
            return redirect('homepage')
    elif role=="admin":
        check_user=login.objects.filter(Q(username=usr_name) & Q(password=userpassword)).first()
        if check_user.password==userpassword:
            request.session['user']=usr_name
            request.session['userid']=check_user.id
            messages.success(request,"Login Success Welcome," , usr_name)
            return redirect ('adminhome')
        else:
            messages.info(request,"Wrong Password Entered")
            return redirect('homepage')
    elif role=="evmanager":
        check_user=login.objects.filter(Q(username=usr_name) & Q(password=userpassword)).first()
        if check_user.password==userpassword:
            request.session['user']=usr_name
            request.session['userid']=check_user.id
            messages.success(request,"Login Success Welcome," , usr_name)
            return redirect ('evmanagerhome')
        else:
            messages.info(request,"Wrong Password Entered")
            return redirect('homepage')
    elif role=="technician":
        check_user=login.objects.filter(Q(username=usr_name) & Q(password=userpassword)).first()
        if check_user.password==userpassword:
            request.session['user']=usr_name
            request.session['userid']=check_user.id
            messages.success(request,"Login Success Welcome," , usr_name)
            return redirect ('technicianhome')
        else:
            messages.info(request,"Wrong Password Entered")
```

```
return redirect('homepage')

else:
    messages.warning(request, "Please    Select    Role")
    return redirect('homepage')

return render(request, 'index.html')
```

Registration page:

```
def homepage(request):
    #return HttpResponse("Welcome to django")

    #if request.method=="POST":
    return    render(request, 'index.html')

def user_registration(request):

    #if    request.method=="post":
    enquiry=registration()
    log=login()

    enquiry.fname=request.POST['fname']
    enquiry.lname=request.POST['lname']
    enquiry.email=request.POST['email']
    enquiry.contact=request.POST['contact']
    enquiry.address=request.POST['address']
    enquiry.city=request.POST['city']
    enquiry.state=request.POST['state']
    enquiry.username=request.POST['username']

    log.username=request.POST['username']
    log.password=request.POST['password']
    log.role='user'
    log.save()

    enquiry.save()
    return render(request, 'index.html')
```

Index page:

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="utf-8">
<meta content="width=device-width, initial-scale=1.0" name="viewport">

<title>AIChargify </title>
<!-- Google Fonts -->
<link
href="https://fonts.googleapis.com/css?family=Open+Sans:300,300i,400,400i,600,600i,700,700i|Jost:300,300i,400,400i,500,500i,600,600i,700,700i|Poppins:300,300i,400,400i,500,500i,600,600i,700,700i"
rel="stylesheet">

<!-- Vendor CSS Files -->
<link href="{% static 'assets/vendor/aos/aos.css' %}" rel="stylesheet">
<link href="{% static 'assets/vendor/bootstrap/css/bootstrap.min.css' %}" rel="stylesheet">
<link href="{% static 'assets/vendor/bootstrap-icons/bootstrap-icons.css' %}" rel="stylesheet">
<link href="{% static 'assets/vendor/boxicons/css/boxicons.min.css' %}" rel="stylesheet">
<link href="{% static 'assets/vendor/glightbox/css/glightbox.min.css' %}" rel="stylesheet">
<link href="{% static 'assets/vendor/remixicon/remixicon.css' %}" rel="stylesheet">
<link href="{% static 'assets/vendor/swiper/swiper-bundle.min.css' %}" rel="stylesheet">
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
<!-- Template Main CSS File -->
<link href="{% static 'assets/css/style.css' %}" rel="stylesheet">
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/font-awesome@6.4.0/css/all.min.css" />

<style>
#hero {
background-image: url(https://png.pngtree.com/background/20230612/original/pngtree-green-forest-with-trees-is-shown-picture-image_3177167.jpg);
background-repeat: no-repeat;
background-size: 100%;
background-position: bottom;
}
</style>
</head>

<body>

<!-- ===== Header ===== -->
<header id="header" class="fixed-top">
<div class="container d-flex align-items-center">

<h1 class="logo me-auto"><a href="index.html">AIChargify</a></h1>
<!-- Uncomment below if you prefer to use an image logo -->
```



```
<!-- <a href="index.html" class="logo me-auto"></a>-->
```

```
<nav id="navbar" class="navbar">
<ul>
<li><a class="nav-link scrollto active" href="#hero">Home</a></li>
<li><a class="getstarted" href="#login" data-bs-toggle="modal" >Login</a></li>
</ul>
<i class="bi bi-list mobile-nav-toggle"></i>
</nav><!-- .navbar -->
```

```
</div>
</header><!-- End Header -->
<div class="modal fade" id="login">
<div class="modal-dialog">
<div class="modal-content">
<div class="modal-header">
<h4 class="text-center w-100">Login Form</h4>
</div>
<div class="modal-body">
<form action="login" method="POST">
{% csrf_token %}
<!-- <div class="form-control bg-green mt-3">Username</div> -->
<input type="text" name="username" class="form-control my-2" placeholder="Username" required/>
<!-- <div class="form-control bg-green mt-3">Password</div> -->
```

```
<div style="position: relative;">
<input type="password" class="form-control"
placeholder="Enter Password" id="pass" name="password" required>
<i class="fa-solid fa-eye" id="togglePassword" onclick="togglePassVisibility()"
style="position: absolute; top: 50%; right: 10px; transform: translateY(-50%);
cursor: pointer; color: #888;"></i>
</div>
```

```
<select name="role" class="form-control my-2">
<option value="" selected>-Select Role-</option>
<option value="user">User</option>
<option value="admin">Admin</option>
<option value="technician">Technician</option>
<option value="evmanager">Ev Manager</option>
</select>
<div class="text-center">
<input type="submit" class="bg-success text-light btn px-3 my-3" value="Login"/>
<!-- <a href="#" class="btn btn-outline-light rounded-pill btn-light bg-green text-decoration-none px-4 py-2
mx-3 mt-3" >Login</a> -->
</div>
</form>
</div>
</div>
</div>
</div>
```

```

<!-- ===== Hero Section===== -->
<section id="hero" class="d-flex align-items-center">

<div class="container">
<div class="row">
<div class="col-lg-6 d-flex flex-column justify-content-center pt-4 pt-lg-0 order-2 order-lg-1" data-
aos="fade-up" data-aos-delay="200">
<h1>AIChargify – Fuel Finder</h1>
<div class="d-flex justify-content-center justify-content-lg-start">
<a href="#register" data-bs-toggle="modal" class="btn-get-started scrollto">Register as User</a>
</div>
</div>
<div class="col-lg-6 order-1 order-lg-2 hero-img" data-aos="zoom-in" data-aos-delay="200">

</div>
</div>
</div>

</section><!-- End Hero -->

```

```

<!-- The Modal-->
<div class="modal" id="register">
<div class="modal-dialog">
<div class="modal-content">

```

```

<!-- Modal Header -->
<div class="modal-header">
<h4 class="modal-title">User - Registration</h4>
</div>

```

```

<!-- Modal body-->
<div class="modal-body">
<form action="user_registration" method="POST">
{% csrf_token %}
<div class="row my-2">
<div class="col">
<input type="text" class="form-control" placeholder="Enter First Name" name="fname" required
pattern="[A-Za-z ]+" title="Only letters and spaces are allowed" />
</div>
<div class="col">
<input type="text" class="form-control" placeholder="Enter Last Name" name="lname" required
pattern="[A-Za-z ]+" title="Only letters and spaces are allowed" />
</div>
</div>
<div class="row my-2">
<div class="col">
<input type="email" class="form-control" placeholder="Enter Email id" name="email" required>
</div>

```

```

<div class="col">
<input type="text" class="form-control" placeholder="Enter Phone No" pattern="[6-9][0-9]{9}"
title="Enter a valid 10-digit phone number starting with 6, 7, 8, or 9" maxlength="10" name="contact"
required>
</div>
</div>
<div class="row my-2">
<div class="col">
<textarea class="form-control" placeholder="Enter Address" required name="address"></textarea>
</div>
</div>
<div class="row my-2">
<div class="col">
<select name="city" class="form-control" required>
<option value="" selected>-Select City-</option>
<option value="Mangalore">Mangalore</option>

</select>
</div>
<div class="col">
<select name="state" class="form-control" required>
<option value="" selected>-Select State-</option>
<option value="Karnataka">Karnataka</option>

</select>
</div>
</div>
<div class="row my-2">
<div class="col">
<input type="text" class="form-control" placeholder="Enter Username" name="username" required>
</div>
<div class="col">

<div style="position: relative;">
<input type="password" class="form-control"
placeholder="Enter Password" id="pass2" name="password" pattern="(?=.*[A-
Z])(?=.*\d)(?=.*[@$!%*?&])[A-Za-z\d@$!%*?&]{6,}" title="Password must contain at least 1 uppercase
letter, 1 number, and 1 special character" required>
<i class="fa-solid fa-eye" id="togglePassword2" onclick="togglePassVisibility2()"
style="position: absolute; top: 50%; right: 10px; transform: translateY(-50%);
cursor: pointer; color: #888;"></i>
</div>
</div>
</div>
<div class="row my-2">
<div class="col">
<input type="submit" value="Register" class="form-control" class="btn btn-primary">
</div>

```

</div>

</form>

</div>

<!-- Modal footer -->

</div>

</div>

</div>

<main id="main">

<!-- ===== Clients Section ===== -->

<!-- ===== About Us Section ===== -->

<section id="about" class="about">

<div class="container" data-aos="">

<div class="section-title">

<h2>About AIChargify</h2>

</div>

{% for msg in messages %}

<div class="alert alert-success alert-dismissible">

<button type="button" class="btn-close" data-bs-dismiss="alert"></button>

{{ msg }}

</div>

{% endfor %}

<div class="row content">

<div class="col-lg-12">

<p>

In the recent decade we have witnessed monumental advancements in electric vehicles and the charging technology. Along with helping cut down on emissions, electric vehicles also have a better power delivery and

prove to be far more efficient as they are able to employ regenerative braking to recharge their batteries while

on the move. Despite their many advantages, electric vehicles still fall short when it comes to aspects such as finding charging stations. Unlike people driving conventional cars, EV owners can't have their vehicles refueled at

any fuel station. Drivers with electric cars have to keep their car charged well in advance before departing.

</p>

</div>

</div>

</div>

</section><!-- End About Us Section -->

```
<div id="preloader"></div>
```

```
<!-- Vendor JS Files -->
```

```
<script src="{% static 'assets/vendor/aos/aos.js' %}"></script>
```

```
<script src="{% static 'assets/vendor/bootstrap/js/bootstrap.bundle.min.js' %}"></script>
```

```
<script src="{% static 'assets/vendor/glightbox/js/glightbox.min.js' %}"></script>
```

```
<script src="{% static 'assets/vendor/isotope-layout/isotope.pkgd.min.js' %}"></script>
```

```
<script src="{% static 'assets/vendor/swiper/swiper-bundle.min.js' %}"></script>
```

```
<script src="{% static 'assets/vendor/waypoints/noframework.waypoints.js' %}"></script>
```

```
<script src="{% static 'assets/vendor/php-email-form/validate.js' %}"></script>
```

```
<!-- Template Main JS File -->
```

```
<script src="{% static 'assets/js/main.js' %}"></script>
```

```
<script>
```

```
function togglePassVisibility2() {
```

```
const password = document.getElementById("pass2");
```

```
const icon= document.getElementById("togglePassword2");
```

```
if(password.type === "password") {
```

```
password.type = "text";
```

```
icon.classList.remove("fa-eye");
```

```
icon.classList.add("fa-eye-slash");
```

```
} else {
```

```
password.type = "password";
```

```
icon.classList.remove("fa-eye-slash");
```

```
icon.classList.add("fa-eye");
```

```
}
```

```
}
```

```
function togglePassVisibility() {
```

```
const password = document.getElementById("pass");
```

```
const icon= document.getElementById("togglePassword");
```

```
if(password.type === "password") {
```

```
password.type = "text";
```

```
icon.classList.remove("fa-eye");
```

```
icon.classList.add("fa-eye-slash");
```

```
} else {
```

```
password.type = "password";
```

```
icon.classList.remove("fa-eye-slash");
```

```
icon.classList.add("fa-eye");
```

```
}
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Admin home page:

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
{% block external_css %}
{% include 'external_css.html' %}
{% endblock %}
<style>
#header{
background: #1754
}

.navbar .getstarted, .navbar .getstarted:focus {
border: 2px solid #198754;
}
.active{
color:#198754!important;
font-weight:bold!important;
}

</style>

</head>

<body>
<!-- ===== Header ===== -->
{% block content %}
{% include 'admin_navbar.html' %}
{% endblock content %}


<!-- ===== Hero Section ===== -->
<section id="" class="d-flex align-items-center">

<div class="container-fluid">
<div class="row">
<div class="col-md-12">
<div class="mt-3 p-5 bg-success text-white text-center rounded">
<h1 class="p-5 ">Welcome {{user_nam}}</h1>

</div>
</div>
</div>
</div>

</section><!-- End Hero -->
```

```

<main id="main">

<div class="container">
<div class="row">
{% for msg in messages %}
<div class="alert alert-warning alert-dismissible">
<button type="button" class="btn-close" data-bs-dismiss="alert"></button>
<strong>{{ msg }}</strong>
</div>
{% endfor %}
<div class="col-md-6">
<div class="card">
<div class="card-header bg-success text-light">Add Area</div>
<div class="card-body">
<form action="adminhome" method="POST">
{% csrf_token %}
<div class="row my-2">
<div class="col">
<select name="area" class="form-control " required>
<option value="" selected>-Select Area-</option>
<option value="Saraswathipuram">Saraswathipuram</option>
<option value="JP Nagar">JP Nagar</option>
<option value="Kuvempunagar">Kuvempunagar</option>
<option value="Vijaynagar">Vijaynagar</option>
<option value="K R Mohalla">K R Mohalla</option>
</select>
</div>
<div class="col">
<input type="submit" class="form-control bg-warning" value="Add Area">
</div>
</div>
</form>
</div>
</div>
</div>
<div class="col-md-6">
<div class="card">
<div class="card-header bg-success text-light">View Added Areas</div>
<div class="card-body">

<table class="table table-striped border">
<thead>
<tr>
<th>Sl No</th>
<th>Name</th>
<th>Uploaded on</th>
</tr>

</thead>
<tbody>

```

```
{% for i in res %}
```

```
<tr>  
<td>{{ forloop.counter }}</td>  
<td>{{ i.name }}</td>  
<td>{{ i.uploaded }}</td>  
</tr>
```

```
{% endfor %}
```

```
</tbody>  
</table>  
</div>  
</div>  
</div>  
</div>  
</div>  
</main>
```

```
{% block external_js %}  
{% include 'external_js.html' %}  
{% endblock %}
```

```
</body>
```

```
</html>
```

EV Manager Home Page:

```
{% load static %}  
<!DOCTYPE html>  
<html lang="en">
```

```
<head>  
{% block external_css %}  
{% include 'external_css.html' %}  
{% endblock %}  
<style>  
#header{  
background: #1754  
}
```



```
.navbar .getstarted, .navbar .getstarted:focus {
border: 2px solid #198754;
}
.active{
color:#198754!important;
font-weight:bold!important;
}
.bookslot{
width: 100%;
border-top: 3px solid #47b2e4;
border-bottom: 3px solid #47b2e4;
padding: 30px;
background: #fff;
box-shadow: 0 0 24px 0 rgba(0, 0, 0, 0.12);

}
.bookslot .form-group{
margin:15px 5px;
}
```

</style>

</head>

<body>

```
<!-- ===== Header ===== -->
{% block evmanagerhome_navbar %}
{% include 'evmanagerhome_navbar.html' %}
{% endblock %}
<!-- End Header -->
```

```
<!-- ===== Hero Section===== -->
<section id="" class="d-flex align-items-center">

<div class="container-fluid">
<div class="row">
<div class="col-md-12">
<div class="mt-3 p-5 bg-success text-white text-center rounded">
```

```
<h1 class="p-5 ">Welcome, EV Manager - {{ evmanager_profile.evmanager }}</h1>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section><!-- End Hero -->
```

```
{% for msg in messages %}
```

```
<div class="alert alert-warning alert-dismissible">
```

```
<button type="button" class="btn-close" data-bs-dismiss="alert"></button>
```

```
<strong>{{ msg }}</strong>
```

```
</div>
```

```
{% endfor %}
```

```
<main id="main">
```

```
<sectionid="contact" class="contact pt-0">
```

```
<div class="container aos-init aos-animate" data-aos="fade-up">
```

```
<div class="row">
```

```
<div class="col-lg-5 d-flex align-items-stretch">
```

```
<div class="info">
```

```
<div class="address">
```

```
<iclass="bi bi-battery-charging"></i>
```

```
<h4>EV Station Name</h4>
```

```
<p>{{ evmanager_profile.stationname }}</p>
```

```
</div>
```

```
<div class="address">
```

```
<iclass="bi bi-person"></i>
```

```
<h4>EV Manger Name</h4>
```

```
<p>{{ evmanager_profile.evmanager }}</p>
```

```
</div>
```

```
<div class="address">
```

```
<iclass="bi bi-map"></i>
```

```
<h4>Area:</h4>
```

```
<p>{{ evmanager_profile.storelocation }}</p>
```

```
</div>
```

```
<div class="address">
```

```
<i class="bi bi-geo-alt"></i>
```

<h4>Location:</h4>

<p>{{ evmanager_profile.storeaddress }}</p>
</div>

<div class="phone">
<i class="bi bi-phone"></i>
<h4>Call:</h4>
<p>{{ evmanager_profile.contact }}</p>
</div>

<div class="phone">
<i class="bi bi-ev-station"></i>
<h4>Charging Slots:</h4>
<p>Slot1 :
{ { evmanager_profile.slot1 } }
{ % if evmanager_profile.slot1 == 'active' %}
Active
{ % else %}
Inactive
{ % endif %}
Slot2 :
{ % if evmanager_profile.slot2 == 'active' %}
Active
{ % else %}
Inactive
{ % endif %}
</p>
</div>

</div>

</div>

<div class="col-lg-7 mt-5 mt-lg-0 d-flex align-items-stretch">
<div class="bookslot">
<div class="card">
<div class="card-header bg-dark text-light">Slots Tracker</div>
<div class="card-body">
<form action="update_slots_tracker" method="POST" class="">
{ % csrf_token %}

```
<input type="text" name="storeid" hidden class="form-control" id="storeid" required="" value="{{
evmanager_profile.storeid }}">
```

```
<div class="form-group">
```

```
<label for="name">Choose Slot</label>
```

```
<select name="charging_slot" class="form-control " required title="Please choose Time Slot">
```

```
<option value="" selected>-Select Charging Slot-</option>
```

```
{% if evmanager_profile.slot1 == 'active' %}
```

```
<option value="slot1">slot1</option>
```

```
{% endif %}
```

```
{% if evmanager_profile.slot2 == 'active' %}
```

```
<option value="slot2">slot2</option>
```

```
{% endif %}
```

```
</select>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="name">Choose Time</label>
```

```
<select name="timeslot" class="form-control" required title="Please choose Time Slot">
```

```
<option value="" selected>-Select Time Slot-</option>
```

```
<option value="8a.m - 9a.m">8a.m - 9a.m</option>
```

```
<option value="9a.m - 10a.m">9a.m- 10a.m</option>
```

```
<option value="10a.m - 11a.m">10a.m- 11a.m</option>
```

```
<option value="11a.m - 12p.m">11a.m- 12p.m</option>
```

```
<option value="12p.m - 1p.m">12p.m- 1p.m</option>
```

```
<option value="1p.m - 2p.m">1p.m- 2p.m</option>
```

```
<option value="2p.m - 3p.m">2p.m- 3p.m</option>
```

```
<option value="3p.m - 4p.m">3p.m- 4p.m</option>
```

```
<option value="4p.m - 5p.m">4p.m- 5p.m</option>
```

```
<option value="5p.m - 6p.m">5p.m- 6p.m</option>
```

```
<option value="6p.m - 7p.m">6p.m- 7p.m</option>
```

```
<option value="7p.m - 8p.m">7p.m- 8p.m</option>
```

```
</select>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="name">Choose Date</label>
```

```
<input type="date" name="bookdate" class="form-control" id="bookdate" required="">
```

```
</div>
```

```
<div class="text-center"><input type="submit" value="Update Slots" class="btn btn-success"/></div>
```

```
</form>
```

```
</div>
```

```
</div>
```

```
<div class="card mt-4">
```

```
<div class="card-header bg-dark text-light">Service Request</div>
```

```
<div class="card-body">
```

```
<form action="store_repair_request" method="post" class="">
```

```
{% csrf_token %}
```

```
<input type="text" name="storeid" class="form-control" id="storeid" required="" hidden value="{{ evmanager_profile.storeid }}">
```

```
<div class="row">
```

```
<div class="col-md-8">
```

```
<div class="form-group">
```

```
<select name="charging_slot" class="form-control " required title="Please choose Charging Slot">
```

```
<option value="" selected>-Select Charging Slot-</option>
```

```
{% if evmanager_profile.slot1 == 'active' %}
```

```
<option value="slot1">slot1</option>
```

```
{% endif %}
```

```
{% if evmanager_profile.slot2 == 'active' %}
```

```
<option value="slot2">slot2</option>
```

```
{% endif %}
```

```
</select>
```

```
</div>
```

```
</div>
```

```
<div class="col-md-4">
```

```
<div class="form-group">
```

```
<div class="text-center"><input type="submit" value="Request to Repair" class="btn btn-success form-control"/></div>
```

```
</div>
```

</div>

</div>

</form>

</div>

</div>

</div>

</div>

<div class="row">

<div class="col-md-3"></div>

<div class="col-md-6">

<div class="card mt-4">

<div class="card-header bg-dark text-light">Request to Repair</div>

<div class="card-body">

<table class="table table-striped border">

<thead>

<tr>

<th>Charging Slot</th>

<th>Service Request on</th>

<th>Request Status</th>

</tr>

</thead>

<tbody>

{% for techni_rep_req in techni_req %}

<tr>

<td>{{ techni_rep_req.charging_slot }}</td>

<td>{{ techni_rep_req.requeston }}</td>

<td>

{% iftechni_rep_req.repair_status == 'Resolved' %}

<div class="badge bg-success"><h6 class="m-0">{{ techni_rep_req.repair_status }}</h6></div>

{% endif %}

{% iftechni_rep_req.repair_status == 'Processing' %}

<div class="badge bg-warning"><h6 class="m-0">{{ techni_rep_req.repair_status }}</h6></div>

{% endif %}

{% iftechni_rep_req.repair_status == 'Pending' %}

<div class="badge bg-info"><h6 class="m-0">{{ techni_rep_req.repair_status }}</h6></div>

{% endif %}

</td>

```
</tr>
```

```
{% endfor %}
```

```
</tbody>
```

```
</table>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
</main>
```

```
{% block external_js %}
```

```
{% include 'external_js.html' %}
```

```
{% endblock %}
```

```
<script>
```

```
// Get the current date
```

```
const today= new Date().toISOString().split('T')[0];
```

```
// Set the min attribute to today's date
```

```
document.getElementById("bookdate").setAttribute("min", today);
```

```
</script>
```

```
</body>
```

```
</html>
```

User home page:

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
{% block external_css %}
{% include 'external_css.html' %}
{% endblock %}
<style>
#header {
background: #1754

}

.navbar .getstarted,
.navbar .getstarted:focus {
border: 2px solid #198754;
}

.active {
color: #198754 !important;
font-weight: bold !important;
}
</style>

</head>

<body>

<!-- ===== Header ===== -->
{% block userhome_navbar %}
{% include 'userhome_navbar.html' %}
{% endblock %}
<!-- End Header -->

<!-- ===== Hero Section===== -->
<section id="" class="d-flex align-items-center">

<div class="container-fluid">
<div class="row">
<div class="col-md-12">
<div class="mt-3 p-5 bg-success text-white text-center rounded">
<h1 class="p-0 ">Welcome <span style="text-transform:capitalize;">{{ user_nam }}</span> (User)</h1>
<h2>Recommended EV Charging Stations (Based on Today's Availability)</h2>
</div>
</div>
</div>
</div>

</section><!-- End Hero -->
```



```

{% for msg in messages %}
<div class="alert alert-success alert-dismissible">
<button type="button" class="btn-close" data-bs-dismiss="alert"></button>
<strong>{{ msg }}</strong>
</div>
{% endfor %}

<main id="main">
<div class="container">
<div class="row">

<div class="col-md-12">
<div class="card">

<div class="card-header bg-success text-light">Search EV Station</div>
<div class="card-body">

<form method="get">
<div class="row my-2">

<div class="col-md-9">
<label for="area">Select Area (Store Location):</label>
<select name="area" id="area" class="form-control" required>
<option value="">--Select Area--</option>
{% for area in areas %}
<option value="{{ area }}" {% if area == selected_area %} selected {% endif %}>{{ area }}</option>
{% endfor %}
</select>
</div>
<div class="col-md-3">
<br>
<button type="submit" class="form-control btn btn-warning">Search</button>
</div>

</div>
</form>

</div>
</div>
</div>

<div class="col-md-1"></div>
<div class="col-md-10"> <br />
<div class="card">

<div class="card-header bg-success text-light">View EV Store</div>
<div class="card-body">
<table class="table table-striped border">
<thead>
<tr>
<th>Sl no</th>
<th>Station Name</th>

```

```

<th>Area</th>
<th>Slots</th>
<th>Congestion Level</th>
<th>Booking Count</th>
<th>Visit</th>
</tr>
</thead>
<tbody>
{% if recommended_stations %}
{% for station in recommended_stations %}
<tr>
<td>{{ forloop.counter }}</td>
<td>{{ station.station_name }}</td>
<td>{{ station.location }}</td>

<td>{{ station.available_slots }}</td>
<td>{{ station.congestion }}</td>
<td>{{ station.booking_count }}</td>
<td>
<form action="{% url 'search_evstation' %}" method="POST">
{% csrf_token %}
<input type="hidden" name="station_id" value="{{ station.station_id }}" required />
<input type="submit" value="Book Now" class="btn btn-success btn-sm" required />
</form>
</td>
</tr>
{% endfor %}
{% else %}
<tr>
<td colspan="7">No stations available at the moment.</td>
</tr>
{% endif %}
</tbody>
</table>
</div>
</div>
</div>
</main>

{% block external_js %}
{% include 'external_js.html' %}
{% endblock %}

</body>

</html>

```

Technician home page:

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
{% block external_css %}
{% include 'external_css.html' %}
{% endblock %}
<style>
#header{
background: #1754

}

.navbar .getstarted, .navbar .getstarted:focus {
border: 2px solid #198754;
}

.active{
color:#198754!important;
font-weight:bold!important;
}

</style>

</head>

<body>

<!-- ===== Header ===== -->
{% block technicianhome_navbar %}
{% include 'technicianhome_navbar.html' %}
{% endblock %}
<!-- End Header -->


<!-- ===== Hero Section===== -->
<section id="" class="d-flex align-items-center">

<div class="container-fluid">
<div class="row">
<div class="col-md-12">
<div class="mt-3 p-5 bg-success text-white text-center rounded">
<h1 class="p-5" style="text-transform:capitalize">Welcome, {{ username }} (Technician) </h1>

</div>
</div>
</div>
</div>
```

```
</section><!-- End Hero -->
```

```
<main id="main">
```

```
<div class="container">
```

```
<div class="row">
```

```
<div class="col-md-12">
```

```
<div class="card">
```

```
<div class="card-header bg-success text-light">Service Request</div>
```

```
<div class="card-body">
```

```
<table class="table table-striped border">
```

```
<thead>
```

```
<tr>
```

```
<th>Store Name</th>
```

```
<th>Area</th>
```

```
<th>Address</th>
```

```
<th>Contact</th>
```

```
<th>Charging Slot</th>
```

```
<th>Service Request on</th>
```

```
<th>Resolved Status</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody>
```

```
<?php
```

```
$loguser=$_SESSION["username"];
```

```
/*
```

```
$query1="select * from ev_station where username='$loguser'";
```

```
$ex=mysqli_query($conn,$query1);
```

```
while($ia=mysqli_fetch_assoc($ex))
```

```
{
```

```
$user_id=$ia["user_id"];
```

```
$fname=$ia["fname"];
```

```
$lname=$ia["lname"];
```

```
}*/
```

```
$query="select * from technician where username='$loguser'";
```

```
$exec=mysqli_query($conn,$query);
```

```
while($i=mysqli_fetch_assoc($exec))
```

```
{
```

```
$technician_id=$i["technician_id"];
```

```
$name=$i["name"];
```

```
}
```

```
$query_tech="select      *      from technician_req      where      repair_status='Processing'      and  
technician_id='$technician_id';  
$exec_tech=mysqli_query($conn,$query_tech);
```

```
while($i_t=mysqli_fetch_assoc($exec_tech))  
{  
$storeid=$i_t["storeid"];  
$repair_id=$i_t["repair_id"];
```

```
$requeston=$i_t["requeston"];  
$charging_slot=$i_t["charging_slot"];  
$repair_status=$i_t["repair_status"];
```

```
$query="select * from ev_station where storeid='$storeid';  
$exec=mysqli_query($conn,$query);
```

```
while($i=mysqli_fetch_assoc($exec))  
{
```

```
$stationname=$i["stationname"];
```

```
$storelocation=$i["storelocation"];  
$storeaddress=$i["storeaddress"];  
$ev_managename=$i["ev_managename"];  
$contact=$i["contact"];
```

```
}
```

```
?>
```

```
{% for tech_req_ in tech_req_data %}  
{% for search_ev in evmanager_profile %}
```

```
{% if search_ev.storeid == tech_req_.storeid %}  
<tr>
```

```
<td>{{ search_ev.stationname }}</td>  
<td>{{ search_ev.storelocation }}</td>  
<td>{{ search_ev.storeaddress }}</td>  
<td>{{ search_ev.contact }}</td>
```

```
<td>{{ tech_req_.charging_slot }}</td>
```

```

<td>{{ tech_req_.requeston }}</td>

<td>
<form action="resolve_service_req" method="post" class="">
{% csrf_token %}
<input type="text" name="storeid" hidden class="form-control" id="storeid" required="" value="{{ tech_req_.storeid }}">
<input type="text" name="repair_id" hidden class="form-control" id="repair_id" required="" value="{{ tech_req_.repair_id }}">
<input type="text" name="charging_slot" hidden class="form-control" id="charging_slot" required="" value="{{ tech_req_.charging_slot }}">
<input type="text" name="tech_id" hidden class="form-control" id="tech_id" required="" value="{{ tech_req_.technician_id }}">

<div class="row">
<div class="col-md-12">
<div class="form-group">
<select name="assign_tech" class="form-control " required title="Please choose Resolved Status">
<option value="" selected>-Choose Resolved Status-</option>

<option value="Resolved" >Resolved</option>
</select>
<input type="submit" value="Resolved Successfully" class="btn btn-success form-control mt-2"/>

</div>
</div>
</div>
</form>
</td>
</tr>
{% endif %}
{% endfor %}

{% endfor %}
</tbody>
</table>
</div>
</div>
</div>

</main>

{% block external_js %}
{% include 'external_js.html' %}
{% endblock %}

</body>

</html>

```

Search EV Station page:

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
{% block external_css %}
    {% include 'external_css.html' %}
{% endblock %}
<style>
    #header{
        background: #1754

    }

.navbar .getstarted, .navbar .getstarted:focus {
    border: 2px solid #198754;
}
.active{
    color:#198754!important;
    font-weight:bold!important;
}
.bookslot{
    width: 100%;
    border-top: 3px solid #47b2e4;
    border-bottom: 3px solid #47b2e4;
    padding: 30px;
    background: #fff;
    box-shadow: 0 0 24px 0 rgba(0, 0, 0, 0.12);

}
.bookslot .form-group{
    margin:15px 5px;
}

</style>

</head>

<body>

    <!-- ===== Header ===== -->
    {% block userhome_navbar %}
    {% include 'userhome_navbar.html' %}
    {% endblock %}
    <!-- End Header -->


    <!-- ===== Hero Section ===== -->
    <section id="" class="d-flex align-items-center">
```

```
<div class="container-fluid">
  <div class="row">
    <div class="col-md-12">
      <div class="mt-3 p-5 bg-success text-white text-center rounded">
        <h1 class="p-5 ">Pre-Book Your EV Slots</h1>
      </div>
    </div>
  </div>
</div>
```

```
</section><!-- End Hero -->
```

```
<section id="contact" class="contact pt-0">
```

```
  <div class="container aos-init aos-animate" data-aos="fade-up">
```

```
    {% for select in select %}
```

```
      <div class="row">
```

```
        <div class="col-lg-5 d-flex align-items-stretch">
```

```
          <div class="info">
```

```
            <div class="address">
```

```
              <i class="bi bi-battery-charging"></i>
```

```
              <h4>EV Station Name</h4>
```

```
              <p>{{ select.stationname }}</p>
```

```
            </div>
```

```
            <div class="address">
```

```
              <i class="bi bi-person"></i>
```

```
              <h4>EV Manger Name</h4>
```

```
              <p>{{ select.evmanager }}</p>
```

```
            </div>
```

```
            <div class="address">
```

```
              <i class="bi bi-map"></i>
```

```
              <h4>Area:</h4>
```

```
              <p>{{ select.storelocation }}</p>
```

```
            </div>
```

```
            <div class="address">
```

```
              <i class="bi bi-geo-alt"></i>
```

```
              <h4>Location:</h4>
```

```
              <p>{{ select.storeaddress }}</p>
```

```
            </div>
```

```
          <div class="phone">
```

```
            <i class="bi bi-phone"></i>
```

```
            <h4>Call:</h4>
```

```
            <p>{{ select.contact }}</p>
```

```
          </div>
```

```
          <div class="phone">
```

```
            <i class="bi bi-ev-station"></i>
```

```
            <h4>Charging Slots:</h4>
```


<p>Slot1 :

```
{% if select.slot1 == 'active' %}
<span class="badge bg-success">Active</span>
{% else %}
  <span class="badge bg-danger">Inactive</span>
{% endif %}
```

Slot2 :

```
{% if select.slot2 == 'active' %}
<span class="badge bg-success">Active</span>
{% else %}
  <span class="badge bg-danger">Inactive</span>
{% endif %}
```

</p>

</div>

</div>

</div>

```
<div class="col-lg-7 mt-5 mt-lg-0 d-flex align-items-stretch">
  <form action="book_slots" method="post" class="bookslot">
    {% csrf_token %}
```

```
    <input type="text" name="userid" hidden class="form-control" required value="{{ usr_profile.id
  }}" >
```

```
    <input type="text" name="storeid" hidden class="form-control" value="{{ select.storeid }}"
  required >
```

```
    <div class="row">
      <div class="form-group col-md-12">
        <label for="name">Name</label>
        <input type="text" name="name" class="form-control" id="name" required="" value="{{
usr_profile.fname }}" >
      </div>
    </div>
```

</div>

```
<div class="form-group">
  <label for="name">Choose Slot</label>
  <select name="chargeslot" class="form-control " required title="Please choose Time Slot">
    <option value="" selected>-Select Charging Slot-</option>
```

```
    {% if select.slot1 == 'active' %}
    <option value="slot1">slot1</option>
    {% endif %}
```

```
    {% if select.slot2 == 'active' %}
    <option value="slot2">slot2</option>
    {% endif %}
  </select>
</div>
```

```

<div class="form-group">
  <label for="name">Choose Time</label>
  <select name="timeslot" class="form-control " required title="Please choose Time Slot">
    <option value="" selected>-Select Time Slot-</option>
    <option value="8a.m - 9a.m">8a.m - 9a.m</option>
    <option value="9a.m - 10a.m">9a.m - 10a.m</option>
    <option value="10a.m - 11a.m">10a.m - 11a.m</option>
    <option value="11a.m - 12p.m">11a.m - 12p.m</option>
    <option value="12p.m - 1p.m">12p.m - 1p.m</option>
    <option value="1p.m - 2p.m">1p.m - 2p.m</option>
    <option value="2p.m - 3p.m">2p.m - 3p.m</option>
    <option value="3p.m - 4p.m">3p.m - 4p.m</option>
    <option value="4p.m - 5p.m">4p.m - 5p.m</option>
    <option value="5p.m - 6p.m">5p.m - 6p.m</option>
    <option value="6p.m - 7p.m">6p.m - 7p.m</option>
    <option value="7p.m - 8p.m">7p.m - 8p.m</option>
  </select>
</div>
<div class="form-group">
  <label for="name">Choose Date</label>
  <input type="date" name="bookdate" class="form-control" id="bookdate" required="">

</div>

```

```

<div class="text-center"><input type="submit" value="Book Slots" class="btn btn-
success"/></div>

```

```

</form>
</div>

```

```

</div>
{% endfor %}

```

```

</div>
</section>

```

```

<script>
// Get the current date
const today = new Date().toISOString().split("T")[0];

// Set the min attribute to today's date
document.getElementById("bookdate").setAttribute("min", today);
</script>
{% block external_js %}
{% include 'external_js.html' %}

{% endblock %}

```

```

</body>

</html>

```

7. TESTING

7. TESTING

7.1 Introduction:

- Software testing is the process of checking computer programs to find out how they work compared to how they're supposed to work. It helps us see how good the software is. To do this, we run the program in different situations—some normal, some unusual—to see how it reacts.
- Testing tries to make the program fail on purpose. This shows us if the software works properly when it should.
- There are two main types of software testing:
- Black box testing: In this method, we focus on what the software should do without worrying about how it's built inside. Tests are based on the software's requirements and functions.
- White box testing: This method involves understanding how the software's code works. We test by looking at the code's statements, branches, paths, and conditions. White box testing is used on small parts, combined parts, and the whole system during the testing process.

7.2 Testing objectives

- Spotting mistakes that programmers might make while creating software.
- Building trust in the software's quality and providing clear information about its standards.
- Stopping mistakes before they happen.
- Ensuring the final product fulfills business needs and user expectations.
- Making sure it aligns with the Business Requirement Specification (BRS) and System Requirement Specification (SRS).

7.3 Testing steps

7.3.1 Unit Testing

Unit testing is when we focus on checking the smallest parts of a software design, which we call modules. In module testing, we test each part separately while the software is still being created. This means programmers run tests on each module to make sure it works correctly and delivers the results we expect. This step is important because it helps us catch any problems early, ensuring that each module functions properly before the entire software is put together..

7.3.2 Integration Testing

- Data can be lost or go wrong when two parts of a program try to communicate. One module might negatively impact another. When small jobs, or sub-functions, are combined, they might not connect properly to perform larger program tasks.
- Integration testing is a methodical way to build and organize a program's framework. It also helps identify mistakes when different segments of a program interact. The purpose is to take modules that have been individually tested and assemble them into the full program structure. After all modules are put together, they undergo testing as a complete unit to ensure everything works correctly.

7.3.3 Validation

- Once integration testing is complete, the software is fully assembled. Any issues with how the parts connect have been identified and fixed. The next step is validation testing, which checks if the software operates as the customer expects. After this testing, there are three potential results:
- The software functions correctly and meets all requirements, so it is accepted.
- It doesn't meet the requirements, and a list of issues is made.
- The new system has undergone validation tests and is confirmed to work well.

7.3.4 Output Testing

After completing validation testing, the next step is to test the outputs of the new system. This is crucial because a system is not useful if it doesn't produce the output we need in the right format. We checked the output format on the screen, and it was correct. This format was planned earlier based on what the user wants. The printed hard copy also matches what the user needs. So, no changes were needed during output testing.

7.3.5 User acceptance testing

The success of any system heavily depends on whether users accept it. To ensure user acceptance, we maintain regular communication with potential users during the development process. This allows us to make changes whenever necessary. We focus on these key areas:

- The design of the input screen.
- The design of the output screen.
- Ensuring online messages are helpful and guide users effectively.

7.3.1.1 Login

Input	Test Condition	Test Output	Comments
Login	If the admin/user /ev manager /technician Username is not entered	Please fill out this field.	Username is required
	If the password is not entered	Please fill out this field.	Password is required
	If the username and password are not valid	User not found.	Enter valid username and password
	If the user /ev manager /technician password are valid.	Successfully logged in.	Appropriate password and email is entered and the user /ev manager /technician is now logged in.
	If the Admin Username and password are valid.	Successfully logged in.	Appropriate password and email is entered and the Admin owner is now logged in to the Admin page.

7.3.1.2 User Registration Form :

Input	Test Condition	Test Output	Comments
Register	If the First name is not entered	Please fill out this field.	Firstname is required
	If the last name is not entered	Please fill out this field.	Lastname is required
	If the city is not entered	Please fill out this field.	City is required
	If the State is not entered	Please fill out this field	State is required
	If the Pincode is not entered	Please fill out this field	Pincode is required
	If the address is not entered	Please fill out this field.	User address is required
	If the email is not entered	Please fill out this field.	User email address is required
	If the phone no. is not entered	Please fill out this field. 10 numeric only	User phone no is required
	If the username is not entered	Please fill out this field.	Username is required
	If the password is not entered	Please fill out this field.	Password required
	If these above fields are valid	Registered successfully	User registered

7.3.2 Integration testing :

7.3.21 Admin Login :

Test No.	Test Description	Expected Output	Expected Output
1	Login	The site is redirected to admin page	Success

7.3.22 User register page and login :

Test No.	Test Description	Test Output	Expected Output
1.	Register	The entered data is stored in database and the site is redirected to login page	Success
2.	Login	The site is redirected to user index page	Success

7.3.23 User Login :

Test No.	Test Description	Expected Output	Expected Output
1	Login	The site is redirected to Customer page	Success

7.3.24 Ev manager/Technician register page and login

Test No.	Test Description	Test Output	Expected Output
1.	Admin create the Ev manager/ Technician	The entered data is stored in admin page and the site is redirected to login page	Success
2.	Login	The site is redirected to Login page	Success

7.3.2.5 Ev manager/Technician login

Test No.	Test Description	Expected Output	Expected Output
1	Login	The site is redirected to admins create Ev manager/technician page	Success

7.3.2.6 Admin Index Page :

Test No.	Test Description	Expected Output	Expected Output
1.	Add area	The site is redirected to the admin page for adding an area	Success
2.	Add Ev station	The site is redirected to the admin page for adding an ev station	Success
3.	Technician Team	The site is redirected to the admin page for hiring an technician	Success
4.	Service request	The site is redirected to Service request page	Success
5.	Completed orders	Site is redirected to completed orders page	Success
6.	Registered users	Site is redirected to registered users page	Success

7.3.2.7 User index Page :

Test No.	Test Description	Expected Output	Expected Output
1.	Search ev station	Site is redirected to userforsearching and booking ev station	Success
2.	View booking	This site is redirected to view booking	Success
3.	View profile	This site is redirected to view profile	Success

7.3.2.8 E V manager index page :

Test No.	Test Description	Expected Output	Expected Output
1.	Charging slot	The site is redirected to ev manager for managing slots and requesting technician for service	Success
2.	User bookings	The site is redirected to admins create traffic police page	Success
3.	Processing orders	The site is redirected to view processing orders	Success
4.	Completed orders	This site is redirected to view completed orders	Success

7.3.2.9 Technician index page :

Test No.	Test Description	Expected Output	Expected Output
1.	Service request	The site is redirected to technician for accepting or rejecting service request	Success

7.3.3 System Testing:

System Testing is the testing of a complete and fully integrated software product. Usually, software is only one element of a larger computer-based system. Ultimately, software is interfaced with other software/hardware systems. System Testing is actually a series of different tests whose sole purpose is to exercise the full computer-based system.

7.3.3.1 System testing tables:

Sl. No	Test condition	Test report
1.	System lading	Successful
2.	System run procedure	Successful
3.	File I/O operation	Successful
4.	Database communication	Successful
5.	Server/client interaction	Successful
6.	Memoryusage	Normal
7.	Systemprocessor usage	Normal
8.	Authentication/Authorization	Successful

8. USER INTERFACE

8. USER INTERFACE

8.1 Screenshots:

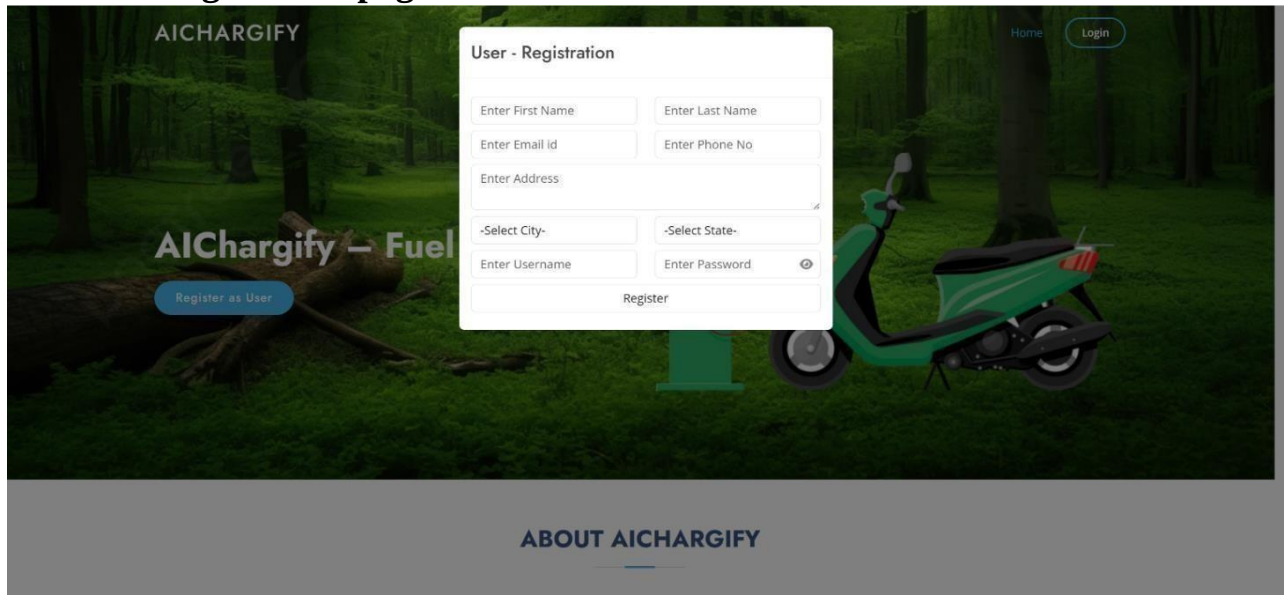
8.1.1 Home page:



ABOUT AICHARGIFY

In the recent decade we have witnessed monumental advancements in electric vehicles and the charging technology. Along with helping cut down on emissions, electric vehicles also have a better power delivery and prove to be far more efficient as they are able to employ regenerative braking to recharge their batteries while on the move. Despite their many advantages, electric vehicles still fall short when it comes to aspects such as finding charging stations. Unlike people driving conventional cars, EV owners can't have their vehicles refueled at any fuel station. Drivers with electric cars have to keep their car charged well in advance before departing.

8.1.2 User registration page :



The image shows the user registration page of the AICHARGIFY website. The background features a green forest scene with a green scooter parked on the right. The AICHARGIFY logo is in the top left, and a 'Home' link and a 'Login' button are in the top right. The main heading is 'AICHargify – Fuel'. Below it is a 'Register as User' button. A white registration form is centered on the page. The form has a title 'User - Registration' and contains the following fields: 'Enter First Name', 'Enter Last Name', 'Enter Email id', 'Enter Phone No', 'Enter Address', '-Select City-', '-Select State-', 'Enter Username', and 'Enter Password' (with a toggle icon). A 'Register' button is at the bottom of the form. At the bottom of the page is a grey bar with the text 'ABOUT AICHARGIFY'.

AICHARGIFY

Home Login

AICHargify – Fuel

Register as User

User - Registration

Enter First Name Enter Last Name

Enter Email id Enter Phone No

Enter Address

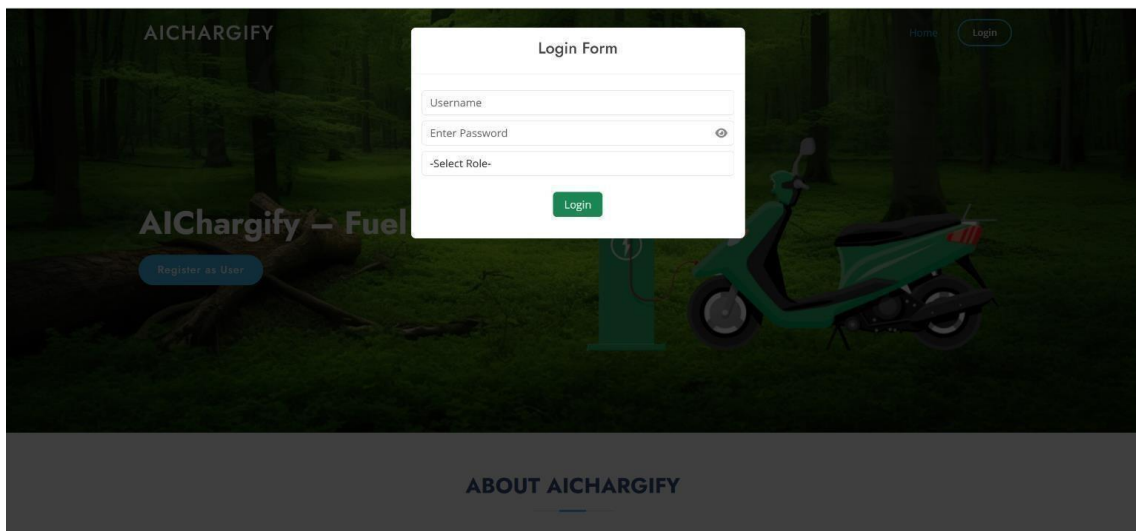
-Select City- -Select State-

Enter Username Enter Password

Register

ABOUT AICHARGIFY

8.1.3 Login page:



The image shows the login page of the AICHARGIFY website. The background features a green forest scene with a green scooter parked on the right. The AICHARGIFY logo is in the top left, and a 'Home' link and a 'Login' button are in the top right. The main heading is 'AICHargify – Fuel'. Below it is a 'Register as User' button. A white login form is centered on the page. The form has a title 'Login Form' and contains the following fields: 'Username', 'Enter Password' (with a toggle icon), and '-Select Role-'. A green 'Login' button is at the bottom of the form. At the bottom of the page is a grey bar with the text 'ABOUT AICHARGIFY'.

AICHARGIFY

Home Login

AICHargify – Fuel

Register as User

Login Form

Username

Enter Password

-Select Role-

Login

ABOUT AICHARGIFY

8.1.4 Admin Home page:

AICHARGIFYAdd AreaAdd EV StationTechnician TeamService RequestCompleted OrdersRegistered UsersLogout

Welcome

Login Success Welcome,

Add Area

-Select Area-

Add Area

View Added Areas

Sl No	Name	Uploaded on
1	Saraswathipuram	April 8, 2025
2	JP Nagar	April 8, 2025

8.1.5 Add EV Station page:

AICHARGIFYAdd AreaAdd EV StationTechnician TeamService RequestCompleted OrdersRegistered UsersLogout

Create and Manage - EV Station

Add EV Store

EV Station Name

EV Manager Name

Enter Phone No

-Select Area-

Enter ev Station address

Enter Username

Enter Password

Add EV Store

View EV Store

Station Name	EV Manager	Contact	Storelocation	Store address	Slot 1 Status	Slot 2 Status	Station Status	Action
Smartfuel	Santosh	9900112255	Saraswathipuram	near fire brigade	active	active	Active	Disable Station
Lightning	Lokesh	8654987566	JP	near Bake point	active	active	Active	Disable Station
smartcharge	vishal	8844552211	JP	near AJ cake world	active	active	Active	Disable Station

8.1.6 Technician Team page:

Create and Manage - Technician Team

Add Technician

Enter Name

Enter Phone No

Enter Username

Enter Password

Add Technician

View Technician

Name	Contact	Action
Akash	9538512139	Delete

8.1.7 Service Request page:

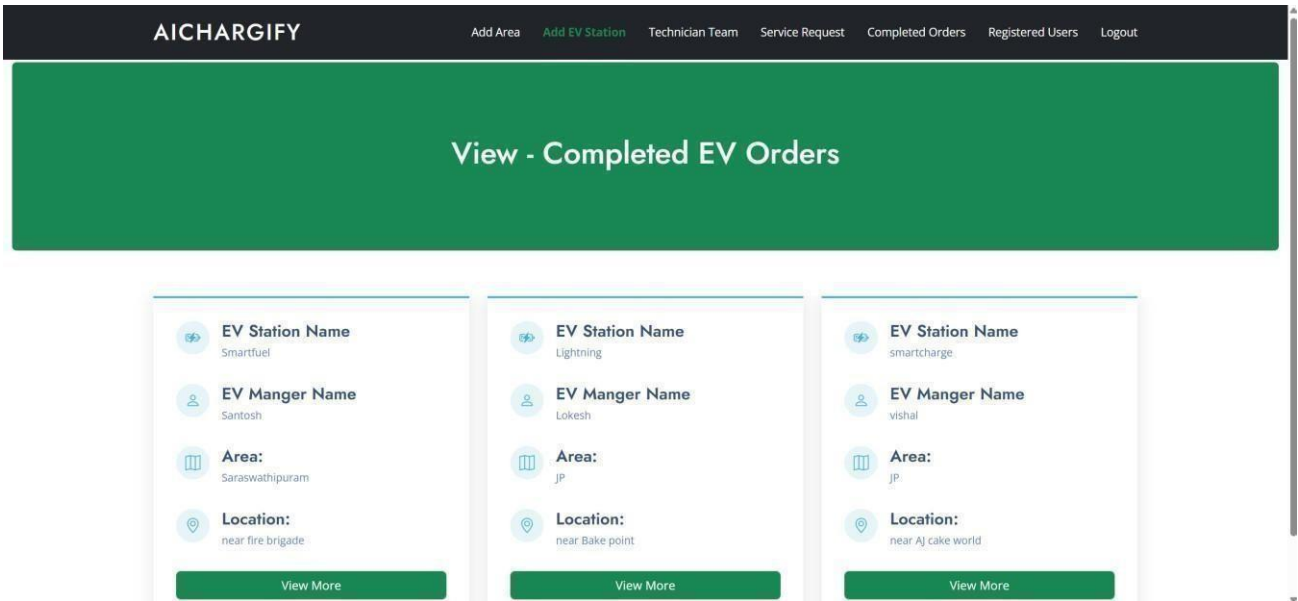
AICHARGIFY

[Add Area](#) [Add EV Station](#) [Technician Team](#) [Service Request](#) [Completed Orders](#) [Registered Users](#) [Logout](#)

View and Mange - Service Request

Add Technician					
Store Name	Area	Charging Slot	Service Request on	Request Status	Forward to Technician
Smartfuel	Saraswathipuram	slot1	April 12, 2025	Pending	-Choose Technician- Assign to Technician

8.1.8 Completed Orders page:



8.1.9 Registered users page :

AICHARGIFY						
Add AreaAdd EV StationTechnician TeamService RequestCompleted OrdersRegistered UsersLogout						
View, Registered Users						
Sl No	Name	Email	Phone no	Address	City	State
1	VINAY MH	vinay123@gmail.com	9657422654	Mangalore	Mangalore	Karnataka
2	Vishal sirsi	Vishal123@gmail.com	8654652446	Sirsi, Karnataka	Mangalore	Karnataka
3	Sudeep k	Sudeepk@123gmail.com	8654223684	Jp Nagar, Bengaluru	Bengaluru	Karnataka

8.1.10 User home page:

AICHARGIFY

[Search EV Station](#)[View Booking](#)[View Profile](#)[Logout](#)

Welcome (User)
Recommended EV Charging Stations (Based on Today's Availability)

Login Success, ✕

Search EV Station

Select Area (Store Location):
--Select Area--

Search

View EV Store

Sl no	Station Name	Area	Slots	Congestion Level	Booking Count	Visit
1	Smartfuel	Saraswathipuram	2	Low Congestion	0	Book Now
2	Lightning	JP	2	Low Congestion	0	Book Now
3	smartcharge	JP	2	Low Congestion	0	Book Now

8.1.11 View booking page:

AICHARGIFY

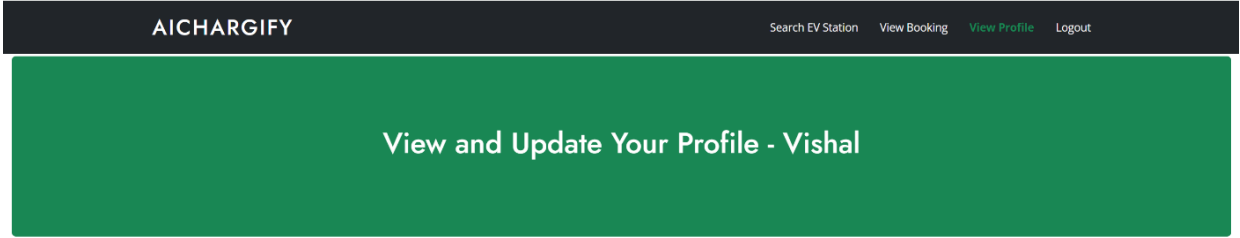
[Search EV Station](#)[View Booking](#)[View Profile](#)[Logout](#)

View Bookings,

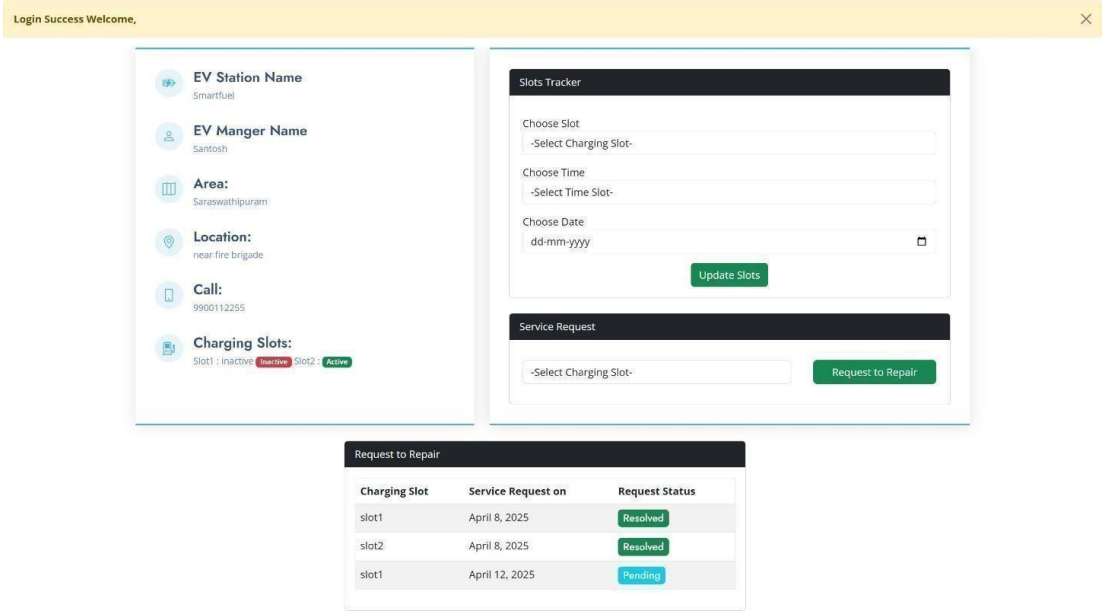
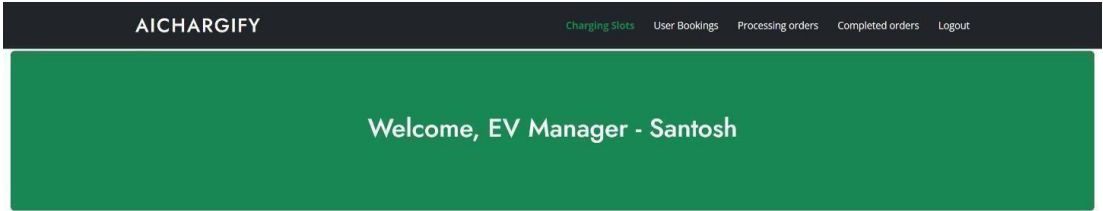
View Bookings

Sl no	Station Name	Area	Booking Date	Charging Slot	Time	Enquired on	Booking Status
1	smartcharge	JP	April 13, 2025	slot1	7p.m - 8p.m	April 12, 2025	Pending

8.1.12 View profile page:



8.1.13 EV manager home page:



8.1.14 User bookings page:

AICHARGIFY

Charging SlotsUser BookingsProcessing ordersCompleted ordersLogout

AIChargify - Charging Request

View Bookings

Sl no	Customer Name	Contact	Booking Date	Charging Slot	Time	Enquired on	Booking Status	Action
2	Vishal sirsi	8654652446	April 13, 2025	slot1	7p.m - 8p.m	April 12, 2025	Pending	-Select Approval Type- Update

8.1.15 Processing orders page:

AICHARGIFY

Charging SlotsUser BookingsProcessing ordersCompleted ordersLogout

AIChargify - Processing Orders

View Bookings

Sl no	Customer Name	Contact	Booking Date	Charging Slot	Time	Booking Status	Action
1	Vishal	8654652446	April 13, 2025	slot1	7p.m - 8p.m	Confirmed	Show this QR Code For Payment  Completed Update

8.1.16 Completed orders page:

AICHARGIFY

Charging SlotsUser BookingsProcessing ordersCompleted ordersLogout

AIChargify - Completed Orders

View Completed Orders

Sl no	Customer Name	Contact	Booking Date	Charging Slot	Time	Booking Status
1	VINAY	9657422654	April 9, 2025	slot1	7p.m - 8p.m	Completed
1	VINAY	9657422654	April 10, 2025	slot1	4p.m - 5p.m	Completed

8.1.17 Technician home page:

AICHARGIFY

Service RequestLogout

Welcome, Akash (Technician)

Service Request

Store Name	Area	Address	Contact	Charging Slot	Service Request on	Resolved Status
Smartfuel	Saraswathipuram	near fire brigade	9900112255	slot1	April 12, 2025	-Choose Resolved Status- Resolved Successfully

9. USER MANUAL

9. USER MANUAL

9.1 Introduction

Electric vehicles (EVs) are transforming the way we think about transportation by making it cleaner and more sustainable. They offer several benefits, such as better energy efficiency, lower running costs, and a reduction in air pollution. Despite these advantages, one significant challenge remains: the limited availability of charging stations. Unlike traditional gas stations, EV charging stations are not yet widespread, which often leads to long waiting times and difficulties in planning travel routes.

To tackle this issue, AIChargify has developed an intelligent, AI-driven system that helps EV users locate nearby charging stations and check their current availability. The platform features an easy-to-use design and offers helpful suggestions, making it simpler for drivers to find places to charge their vehicles while on the go. This not only enhances the convenience of traveling with EVs but also encourages more people to adopt this eco-friendly mode of transportation.

9.2 Hardware Requirements:

- Processor: Intel core i3 or higher
- RAM: 4GB and above
- Hard disk Utilization: 4GB and above
- Input Devices: Mouse, Keyboard
- Output Devices: Monitor, Printer

9.3 Software Requirements:

- Operating system: Windows
- Editor: Visual Studio Code
- Database (Back End): MySQL
- Server :Apache
- Browsers: Firefox, Chrome or anyother browser

10. CONCLUSION

10. CONCLUSION

We designed AI Chargify Fuel Finder using the requirements we collected and the best technology available. This platform helps connect charging stations, users, administrators, and technicians in an interactive way. Right now, there isn't an app to manage bookings for electric vehicles (EVs). People usually have to search for EV charging stations by themselves, which can be time-consuming. With AI Chargify Fuel Finder, everyone—whether they need to charge their car, manage the stations, or fix technical issues—can come together in one place. The platform provides easy access and solutions for all, making the process much more efficient for everyone involved.

Future scope:

- Future scope of this project can be planned that to host it on a real server and increase the maximum usage. In Future,
- we also integrate SMS integration and Email so that instead of login and check notification every time we get notification through sms or by email on every interaction,
- we will integrate a payment gateway then they can pay through the payment gateway

11. BIBLIOGRAPHY

11. **BIBLIOGRAPHY**

11.1 **web references:**

- <https://www.google.co.in/>
- <https://kineticgreenvehicles.com/blog/can-we-charge-electric-scooter-at-home/>
- <https://www.cardekho.com/electric-charging-station>
- https://en.wikipedia.org/wiki/Charging_station
- https://en.wikipedia.org/wiki/Web_application
- <https://en.wikipedia.org/wiki/Database>
- <https://en.wikipedia.org/wiki/PHP>
- <https://www.apachefriends.org/>
- https://developer.mozilla.org/en-US/docs/Learn/CSS/CSS_layout/Responsive_Design#:~:text=Responsive%20web%20design%2C%20or%20RWD,technology%20%E2%80%94%20it%20is%20an%20approa

12. PLAGIARISM REPORT



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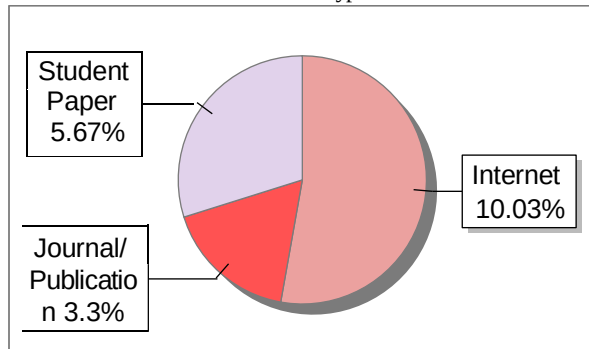
Author Name	Vinay MH
Title	AI chargify
Paper/Submission ID	3559503
Submitted by	librariansrinivasuniversity@gmail.com
Submission Date	2025-04-29 17:31:28
Total Pages, Total Words	77, 7658
Document type	Project Work

Result Information

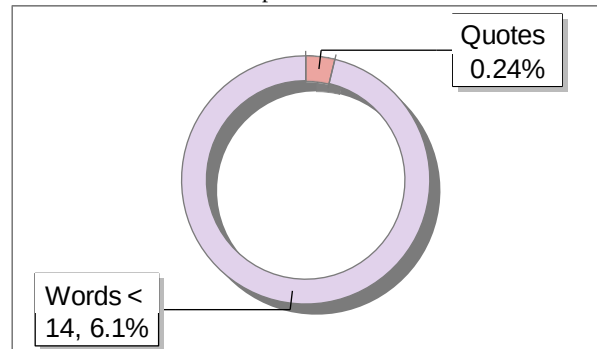
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19	amcdigitalbd.com	<1	Internet Data
20	Submitted to Visvesvaraya Technological University, Belagavi	<1	Student Paper
21	qdoc.tips	<1	Internet Data
22	www.scribd.com	<1	Internet Data
23	www.scribd.com	<1	Internet Data
24	www.almabetter.com	<1	Internet Data
25	Appendix A Setting KPIs and Targets Using a Ten-Step Process by -2020	<1	Publication
26	cdc.gov	<1	Internet Data
27	docplayer.gr	<1	Internet Data
28	docplayer.net	<1	Internet Data
29	geeksarticle.com	<1	Internet Data
30	pdfcookie.com	<1	Internet Data
31	www.crazygames.com	<1	Internet Data
32	documents.mx	<1	Internet Data
33	qdoc.tips	<1	Internet Data

34	repositorio.unal.edu.co	<1	Internet Data
35	sdbj.com	<1	Internet Data
36	Chaucer and the Subversion of Form ed by Thomas A Prendergast and Jessica Rose by Cowdery-2019	<1	Publication
37	docplayer.net	<1	Internet Data
38	docs.openvino.ai	<1	Internet Data
39	forum.effectivealtruism.org	<1	Internet Data
40	moam.info	<1	Internet Data
41	newprairiepress.org	<1	Publication
42	OUTROS ESPAOS, NOVOS SABERES a atuao do pedagogo no Centro de Referncia e By Keyla Cardoso Tavares, Kssia, Yr-2017,3,9	<1	Publication
43	Phased development of high-concentration photovoltaic modules by W-1993	<1	Publication
44	REPOSITORY - Submitted to Kuvempu University on 2025-04-25 17-11 3534960	<1	Student Paper
45	signeasy.com	<1	Internet Data
46	universitykart.com	<1	Internet Data
47	www.lss-helpdesk.com	<1	Internet Data
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49	www.readbag.com	<1	Internet Data
50	Construction and beam test of the ZEUS forward and rear calorimeter by A-1991	<1	Publication

51	docplayer.net	<1	Internet Data
52	edanz.com	<1	Internet Data
53	journals.library.columbia.edu	<1	Publication
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